

Project Report On:

TrustVote: Online Voting using BlockChain

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Under the Guidance of
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[B. Tech. Information Technology]

At Department of Information Technology

[2024-2025]

At



Vishwakarma Institute of Information Technology,

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Affiliated To



Savitribai Phule Pune University,

Chapter I

Certificate From Institute



This is to certify that the work entitled “**TrustVote: Online Voting using BlockChain**” is a bonafide work carried out by **Tanish Bagal, Harshal Dahiwal, Shrinivas Kathare, Shreyas Raijade**, in partial fulfillment of the award of Bachelor of Technology in Information Technology, Savitribai Phule Pune University, Pune, during the year 2021. The project report has been approved as it satisfies the academic requirements in respect of the project work prescribed for the Bachelor of Technology Degree.

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Project Guide

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Head, IT Department

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Date:

Examiners: 1.

2.

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Chapter II: Industry Support Letter



Vijay Computer Data System

Quality and Time matters

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Date: 30 July, 2024

Subject: Provision of Industrial support for project

We are delighted to write this letter in support of the final year project undertaken by the students

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Representing Vijay Computer Data system, I have had the pleasure of working closely with these students on their project titled "TrustVote Online Voting using blockchain" and I have been thoroughly impressed by their dedication and innovation.

At Vijay Computer Data system, we are passionate about fostering innovation and supporting the educational journeys of aspiring professionals in technology. Throughout this project, we have provided the students with valuable industry insights, technical guidance, and necessary resources to help them succeed.

The students have shown exceptional commitment, creativity, and technical skills in addressing the challenges of their project. Their work on blockchain technology not only aligns with industry standards but also demonstrates significant potential for practical applications. Our team has been genuinely impressed by their ability to combine theoretical knowledge with practical application, their collaborative spirit, and their determination to deliver a high-quality solution. We believe the skills and experiences they have gained through this project will be immensely beneficial as they transition into their professional careers. In support of their efforts, Vijay Computer Data has provided mentorship, equipment, software, funding.

We eagerly anticipate the final outcomes of their project and look forward to continuing our support for such initiatives in the future. Thank you for your attention and support for these talented students.

Sincerely,

Asha Kadam
Founder
Vijay Computer Data System

Chapter III: Project Completion Letter



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Date: 14 November 2024

To,

The Head of Department
Information Technology Department
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Subject: Confirmation of Project Completion – TrustVote: Online Voting using Blockchain

Respected Sir/Madam,
We are pleased to inform you that,

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students in the Information Technology Department at Vishwakarma Institute of Information Technology (VIIT), has successfully completed their project titled "**TrustVote: Online Voting using Blockchain**" with the support of *Vijay Computer Data System*

Throughout the project, students demonstrated excellent technical proficiency, problem-solving skills, and a thorough understanding of blockchain technology, which played a crucial role in developing an innovative and secure online voting solution. Their commitment to achieving the project objectives reflects positively on their ability to handle complex technical challenges, especially in the critical area of secure online voting.

We believe this project has provided students with valuable real-world experience and essential skills that will benefit their future academic and professional pursuits. It has been a pleasure to collaborate with Vishwakarma Institute of Information Technology (VIIT) and support the skill development of its students.

Sincerely,

Asha Kadam
Founder
Vijay Computer Data System

Chapter IV

Acknowledgement

We take this opportunity to thank Head of the Department **Prof. Pravin Futane** and our project guide **Prof. Shalini V. Wankhede** for their valuable guidance and providing all the necessary facilities, which were indispensable in the completion of this project report. We are also thankful to all the staff members of the Department of Information Technology, VIIT, Pune for their valuable time, support, comments, suggestions and persuasion. We would also like to thank **Mrs.Suruchi Dedgaonkar, Mr.Mandar Mokashi**, and the institute for providing the required infrastructure, internet and library facility.

Tanish Bagal
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Chapter V

Abstract

This project report presents the design and implementation of a Decentralized voting system using blockchain technology. The purpose of this system is to provide a secure, transparent, and immutable platform for conducting elections. The system employs the use of smart contracts on the Ethereum blockchain to automate the voting process and ensure that all votes are counted accurately. The system was developed using Solidity, a programming language for smart contracts, and Truffle, a development framework for Ethereum. The system's frontend was developed using HTML, CSS and JavaScript. Nodejs ,Expressjs and Mongoddb for system's backend. The system's security was ensured through the use of encryption techniques and private keys to secure users' identities and their voting records. The system's architecture allows for the registration of voters, the creation of ballots, the casting of votes, and the tallying of results, all on the blockchain. The system's decentralized nature ensures that votes cannot be tampered with, and the results are transparent and available for all participants to verify. This project report concludes that a decentralized voting system using blockchain technology has the potential to revolutionize the way elections are conducted, by providing a secure, transparent, and immutable platform that can be trusted by all participants. The system can be used in various settings, including governmental and non-governmental elections, corporate boardroom elections, and more.

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Abbreviations:

- OVS : Online Voting System
- BVS : Blockchain Voting System
- SC : Smart Contract
- UI : User Interface
- DB : Database
- ETH : Ethereum
- API : Application Programming Interface
- DApp : Decentralized Application
- PKI : Public Key Infrastructure

Chapter VI

INTRODUCTION

1.1 MOTIVATION

The inspiration behind a blockchain-based online voting initiative originates from the need to build a more secure, transparent, and trustworthy electoral process. Transparency is a common worry in traditional voting systems since voters have little information about whether their vote was correctly counted or whether there was no election manipulation. This problem can be solved by blockchain technology, which ensures that every vote is permanently and publicly recorded. This way, voters' privacy is protected and anybody can independently confirm the correctness of the election results. Security is another significant driving element. Voting systems are vulnerable to fraud, manipulation, and cyberattacks, particularly in online settings. Blockchain technology secures votes via decentralized consensus methods and cryptographic algorithms, making it nearly impossible for bad actors to tamper with the voting process or change the results. Election integrity must be preserved at all costs, particularly in areas where voter manipulation is a prevalent worry. Democratic systems require confidence in the voting process to be functional. By guaranteeing transparency and security, blockchain technology contributes to the development of that trust by enabling voters to verify that their vote is counted and that there is no chance of fraud or manipulation. Additionally, the effectiveness of blockchain technology minimizes delays and the possibility of disputes over election results by enabling quicker vote counting and quicker notification of results.

1.2 NEED FOR ONLINE VOTING

A number of variables that emphasize the drawbacks of conventional voting systems and the possible advantages of a digital approach have made online voting more and more necessary in today's society. One of the key needs is convenience and accessibility. Conventional voting procedures frequently call for voters to physically show up at polling places, which can be difficult for those who live in

rural areas, are elderly, disabled, or have limited time. Voting online lowers the obstacles to participation and may boost turnout by enabling people to cast their ballots from any location. The voting procedure's effectiveness is another important factor. Traditional systems often include manual counting, significant waiting times, and the need for extensive logistical assistance, which can contribute to delays in announcing results and increased expenses. By automating the counting of votes, facilitating speedier announcements of the results, and eliminating the need for physical infrastructure, online voting streamlines the procedure and eventually lowers expenses. Additionally, there is the issue of openness and security. Online voting systems, particularly those utilizing blockchain technology, offer a better level of security through encryption and decentralized verification techniques, whereas traditional voting systems can be subject to fraud, manipulation, and tampering. By guaranteeing that each vote is precisely and irrevocably counted, these technologies can lower the possibility of electoral fraud and increase voter confidence. Additionally, expectations for contemporary, digital voting alternatives have increased due to the global shift towards digitalization in numerous industries, such as banking, retail, and communication. A modern voting system that takes into account this digital change is becoming more and more necessary as societies become increasingly interconnected and dependent on technology. In the digital age, online voting provides a future-proof, scalable option that meets the changing needs of citizens. In general, the desire for a more transparent, safe, efficient, and inclusive election process that can adjust to the needs of contemporary society is what motivates the need for online voting.

1.3 BRIEF INTRODUCTION OF TRUSTVOTE

Democratic voting is such an important and significant event in every country, and the current way we vote usually involves either ballot papers or EVMs. While these methods are in place, they do come with some challenges, like transparency issues, lower voter participation, vote tampering, a lack of trust in the election authority, problems with unique voter IDs, delays in announcing results, and, most

importantly, security concerns. When we think about implementing a digital voting system, the security of digital voting really stands out as a primary concern. Given the enormous decisions involved, we want to be absolutely sure that the system can protect data and guard against possible attacks. One potential solution to these security challenges could be the use of blockchain technology. Blockchain technology opens up countless possibilities. It's a distributed ledger technology that enables digital assets to be exchanged in a peer-to-peer decentralized network. This advancement in distributed ledger technology is truly exciting! A block is essentially a collection of all the transactions that have taken place. Blockchain comes with fantastic features like immutability, decentralization, security, transparency, and anonymity. Combining blockchain with smart contracts presents a wonderful opportunity for creating a safer, more secure, and transparent e-voting system. In this paper, we've worked on and tested a sample e-voting application as a smart contract for the Ethereum network, utilizing blockchain technology through wallets and the Solidity programming language.

1.4 REASONS BEHIND TRUSTVOTE

The necessity to get beyond the serious drawbacks of conventional voting procedures is what spurred the creation of a blockchain-based electronic voting system. Voter fraud is a major factor, as it has the potential to seriously damage elections' credibility. Conventional systems are susceptible to manipulation, tampering, and other types of intervention that jeopardize the accuracy of the outcomes. This project uses blockchain technology to establish an immutable, secure, decentralized voting system where votes cannot be changed or falsified after they are cast. This improves the process's overall security. Another important component driving this project is transparency. Voters frequently have little to no visibility into the handling, tallying, and verification of their ballots during election procedures. The public's confidence in the system may be weakened by this lack of openness. A blockchain-based solution overcomes this by establishing a public record that assures every vote can be monitored and confirmed without revealing

voter identities. This offers a degree of transparency that is challenging to accomplish with conventional techniques. Blockchain's efficiency is another important factor. Conventional voting procedures can be laborious and sluggish, especially in major elections. This causes delays in the announcing of results and the counting of votes. Blockchain-based electronic voting can automate the tallying of votes, resulting in quicker results and a lower chance of human mistake. This improves the process's dependability and efficiency, which is crucial in election situations where there is a deadline. Additionally, since traditional voting methods are frequently the target of hackers, security is a big worry. Because of blockchain's decentralized architecture, hackers have a much tougher time interfering with or controlling the process. Through the use of cryptographic algorithms, every vote is safeguarded, guaranteeing that only legitimate votes are counted and that they are kept unaltered throughout. The requirement to increase confidence in election procedures serves as another driving force. The legitimacy of democratic institutions depends on the public's belief in the impartiality and accuracy of elections. Voter trust can be restored and strengthened by using an electronic voting system built on blockchain, which offers a way to guarantee that the process is transparent and impervious to tampering. Additionally, elections may be more inclusive due to the accessibility and ease of internet voting.

Chapter VII

LITERATURE SURVEY

2.1 LITERATURE REVIEW

Id	Year, Title, Author	Methodology	Limitation	Comparison to Current Study
[1]	ICMPC 2023 Decentralized and Automated Online Voting System using Blockchain Technology Polepaka Sanjeeva ^{1*} , M. Sai Sathwik ¹ , G. SaiPrasad ¹ .	Blockchain Integration: The proposed online voting system uses Ethereum Blockchain to provide a decentralized ledger for recording votes. Smart Contracts: Smart contracts on the Ethereum Blockchain automate the voting process, including ballot counting and result verification. Facial Recognition: Voter authentication is performed through facial recognition technology. This biometric verification ensures that only eligible voters can participate and prevents fraudulent voting by verifying voter identity against stored data. Voting Process: Voters access the system, undergo facial recognition for authentication, and cast their votes. authentication allows voters to select	1.Facial Recognition Accuracy: Facial recognition systems may have limitations in terms of accuracy and reliability, particularly under varying lighting conditions or with diverse demographic groups. 2.Manipulation Risks: Online voting systems may be susceptible to coercion or vote buying. Designing mechanisms to detect and prevent these practices is essential to maintain the fairness of elections.	The current study compares the implementation of the existing blockchain voting system comprising three levels: authentication, voting, and counting.

		candidates from a district-specific list. Vote Verification: Votes are verified by multiple miners who ensure that only valid votes are added to the public ledger. Each vote is processed as a separate transaction, and malicious votes are rejected. Transaction Tracking: Voters receive notifications with transaction IDs for tracking their votes. This system ensures that voters can verify their participation while maintaining confidentiality.	3.Data Exposure: While Blockchain provides transparency, it can also expose voter identities and preferences. Balancing transparency with privacy is crucial, and techniques like zero-knowledge proofs or advanced encryption methods may be required to address this issue.	
[2]	2020 A Secure Digital Voting System based on Blockchain Technology S. S. Bhalerao, et al.	Proposed a secure digital voting system using blockchain technology, emphasizing the system's architecture, security features, and implementation details.	Focused primarily on the technical aspects, with limited discussion on scalability and user accessibility.	This study builds on the concept by addressing additional aspects such as scalability and practical deployment in large-scale elections.
[3]	2018 Blockchain-Based E-Voting System A. Kiayias, et al.	Implemented a blockchain-based e-voting system, focusing on privacy, verifiability, and resistance to tampering.	Lacked practical deployment analysis and did not consider large-scale election scenarios.	The current study compares the implementation strategies and enhances the security protocols for better real-world applicability.

[4]	June 2016 Electronic Voting Service Using Block-Chain Kibin Lee, Joshua I. James, Tekachew Gobena Ejeta, Hyoung Joong Kim	The paper introduces a novel voting system that leverages blockchain technology to enhance security and transparency. It involves several steps: filtering transactions, implementing a consensus protocol, and using a decentralized ledger to record votes. The system incorporates a trusted third party (TTP) for additional validation and employs blockchain for immutability.	The system's effectiveness depends on the accurate functioning of the TTP and the blockchain protocol. Challenges include potential vulnerabilities in the transaction filtering process, maintaining consensus in a decentralized network, and addressing scalability issues. Additionally, the system's dependence on the TTP could introduce points of failure or bottlenecks.	This study Maintains redundant copies of the blockchain data to prevent data loss and ensure continuity in case of an attack or failure.
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[5]	2019 Blockchain And Finger Print Enabled E-Voting Aravind, Gokul Raj, Mohanraj	Proposed system Fingerprint Authentication for Identity Verification due to faster and better than other biometric data. They have used Blockchain Data Storage and Homomorphic encryption. Also the system has two types of users one is	The paper mentions that no full implementation of blockchain-based e-voting has occurred for a national election. Although the proposed system aims to	The proposed system aims to decentralize the voting process, potentially increasing security and transparency.
		Election Officer & another is Booth Manager, Booth Manager System developed with voter's functionality where voters are going to poll.	reduce costs associated with traditional voting methods, the complexity of implementing this may present significant challenges.	

[6]	2023 Transparent voting system using blockchain V. Anitha ,*, Orlando Juan Marquez Caro , R. Sudharsan S. Yoganandan M. Vimal	The methodology of the research paper involves creating a blockchain-based voting system to ensure secure and transparent elections. It utilizes the Ethereum blockchain and smart contracts to automate the voting process, including voter authentication and vote tallying, eliminating the need for intermediaries. The system ensures voter anonymity through cryptographic techniques and provides real-time election results to enhance transparency.	Complex for Non-Tech Users: Users unfamiliar with blockchain may find it difficult to use digital wallets like MetaMask. Scalability Issues: The system might slow down with a large number of votes due to Ethereum's transaction limits. Cost of Transactions: Each vote incurs a fee on the Ethereum network, which could be costly in large elections.	The current study compares the implementation of the existing blockchain voting system comprising three levels: authentication, voting, and counting.
[7]	June 2024	The study describes a blockchain-based electronic voting system in which the system uses homomorphic	The system's scalability is hampered by the difficulty of handling many	By employing more effective consensus techniques, streamlining the

	Online Voting Platform Using Blockchain Technologies Sonali Rangdale, Nagesh Raykar, Santosh Borde, Prashant Kumbharkar, Udit kumar	encryption and the cryptographic hash algorithm Keccak-256 to enable voter identity verification, safe ballot issuance, and encrypted vote storage. Voters may monitor and verify their ballots while preserving data integrity and privacy thanks to the decentralized Ethereum blockchain used to store votes. The most recent votes are counted, and the mechanism is meant to avoid duplicate voting.	votes at once, particularly in areas with big populations. Additionally, its performance and accessibility may be restricted by the requirement for technical know-how, substantial device resources, and reliable internet connectivity.	blockchain for increased performance, and developing user-friendly interfaces with offline voting choices for places with inadequate internet connectivity, we may get around these problems. Furthermore, increasing technological accessibility can be achieved through user education.
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2.2 Review of Existing Systems

During the brainstorming phase for system requirements and functionalities of our Online Voting System using Blockchain, we explored various existing online voting and election management systems. One of the most well-known systems is ElectionBuddy, which provides secure and anonymous voting for organizations. It offers features like vote tracking, audit logs, and customizable ballots. However, while ElectionBuddy focuses on ease of use and security, it does not employ blockchain technology for immutability and decentralized trust, which are key advantages in our proposed system.

Another system we reviewed was Voatz, a mobile voting platform used in some governmental elections. While Voatz does incorporate blockchain for vote recording, its reliance on mobile technology has raised concerns regarding security vulnerabilities, such as those related to mobile device integrity.

The key disadvantages of existing systems include:

- **Lack of Full Transparency:** Many traditional systems do not allow voters to verify that their vote was recorded as intended in a publicly transparent way.
- **Centralization Risks:** Centralized databases are vulnerable to tampering or hacking, unlike a blockchain-based decentralized system.
- **Voter Privacy Concerns:** Some systems may not fully ensure voter privacy, with potential for unauthorized access to voting records.

In contrast, our Blockchain Voting System seeks to address these shortcomings by:

- Providing a fully decentralized and immutable platform using blockchain, where every vote is transparently recorded and cannot be tampered with.
- Ensuring voter privacy and security through encryption techniques and the use of private keys.
- Offering real-time transparency, allowing all participants to verify election results.
- Supporting features like voter registration, secure login, vote casting, and result viewing—all while maintaining trustless integrity.

Our system supports two types of user roles:

- **Voters:** Voters can access the platform to register, log in, cast their votes, and view election results.
- **Admin:** The admin has access to voter management, candidate management, and real-time monitoring of the election results, as well as control over starting and closing elections.

By building on blockchain technology, our system offers a robust, secure, and transparent alternative to traditional voting platforms.

2.2.1 System Functions

Voter Functions:

- **Register:** Sign up for the voting system using personal credentials (Aadhar number).
- **Login:** Access the system with valid credentials.
- **View Candidate List:** See all the candidates for the election.
- **Cast Vote:** Submit a vote for the chosen candidate.
- **View Voting Status:** Confirm that the vote has been successfully cast.
- **View Election Results:** Access the final voting results once the election is concluded.
- **Email Confirmation:** Receive email verification after casting a vote.
- **Edit Profile:** Update personal information within the system.
- **Logout:** Securely log out from the system.

User (General Voter Data and Election Info):

- **Get User Information:** Access details of voters who have participated in the election.
- **View Election Progress:** Track real-time voting progress and results.
- **Election History:** View past election results and voting records.
- **Voting Transparency:** Verify the integrity of votes cast by accessing the public ledger.

Admin Functions:

- **Add/Remove Candidates:** Manage the list of candidates for elections.
- **View Registered Voters:** Access a list of all voters registered in the system.
- **Manage Elections:** Start and close election processes.
- **View Voting Results:** Access and monitor the voting results in real time.
- **Update System Settings:** Modify system settings like voting periods or candidate data.
- **Audit Voting Records:** Review blockchain logs for transparency and integrity.
- **Add/Update Voter Credentials:** Add or update credentials for voters if required.

Chapter VIII

PROJECT STATEMENT

3.1 PURPOSE BEHIND THE PROJECT

The blockchain-based electronic voting project aims to solve the drawbacks of conventional voting procedures by developing a safe, open, and effective election system. Through the use of blockchain technology, the initiative seeks to improve election integrity by thwarting voter fraud and manipulation. The initiative aims to guarantee that every vote is precisely recorded and verifiable by offering a decentralized and impenetrable platform. This can greatly increase public confidence in the political process. The concept also aims to increase voter convenience and accessibility. By enabling voters to cast ballots from any location, online voting lowers potential obstacles to participation, such as geographic distance or physical limitations. The goal of this improved accessibility is to encourage greater participation in the political process and voter turnout. Easing the voting procedure is another important goal. Long-drawn out processes are frequently used in traditional systems to announce results and count votes, which can cause delays and annoyance. The initiative seeks to automate these procedures using blockchain in order to minimize human error and produce faster outcomes. The project's ultimate goal is to modernize voting, making it more secure, transparent, and inclusive in the process, strengthening the fundamentals of democratic governance and participation.

3.2 DECISION OF SCOPE

A blockchain-based electronic voting project's main goal is to provide a safe, open, and effective voting system that overcomes the drawbacks of conventional approaches. The project's primary goal is to improve election integrity by guarding

against voter fraud and tampering and making sure that votes are correctly counted and validated via a decentralized blockchain system. It is crucial to consider both the technical and practical issues while choosing the scope. Clearly defining the project's main goals is the first step. The system is made to deliver dependable and quick results while enhancing security, accessibility, and transparency. The system's design choices and functions will be influenced by these goals. The system's intended audience is yet another important factor to take into account. Voters, election officials, and other parties involved with the platform's interaction are included in this. A project that is successful will consider these users' demands and make sure the system is user-friendly and accessible, even for non-techies. Making the UI user-friendly is crucial given the variety of users, since this will enable smooth voting participation. Additionally, the geographic scope must be specified precisely. The technical and regulatory requirements will vary depending on whether the system is used for municipal, regional, or nationwide elections. Smaller pilots might be easier to manage as a first step, but larger-scale deployments might call for a more intricate architecture. Determining the scope technically primarily revolves around choices made about the blockchain platform, encryption strategies, and user authentication mechanisms. Whether to use a public or private blockchain will rely on how much control and transparency you need over the system. A crucial factor to take into account early in the project is figuring out how the system will interact with the current electoral infrastructure, such as voter databases and election management tools. Making ensuring the new platform enhances current procedures rather than interferes with them is the aim. To make sure the system performs as planned, the project will require a comprehensive testing and validation plan. This entails thorough security testing to stop weaknesses and pilot projects to spot possible usability problems.

Finally, it's critical to manage the project's budget and schedule. The resources at hand will determine the scope, and careful planning is required to strike a balance between functionality and limitations like time and money. The project will be more likely to stay on course and reach its goals if there is a clear timetable for

development, testing, and deployment. A clear set of technical, user-focused, and regulatory components are included in the scope of the blockchain-based electronic voting project. These components are all intended to produce a safe, open, and easily accessible voting environment. Through meticulous evaluation of these variables, the project can be structured to efficiently accomplish its goals while tackling the obstacles presented by conventional voting methods.

3.3 METHODOLOGY

3.3.1 Proposed System Architecture

The proposed decentralized online voting system is built using a three-layered architecture, consisting of the frontend, backend, and blockchain. The frontend is developed using HTML, CSS, and JavaScript, providing a user-friendly interface for voter registration, login, and vote casting. The backend, developed with Node.js and Express.js, handles server-side operations, including authentication, data validation, and communication with the blockchain. MongoDB is used to store non-sensitive user data such as login credentials.

The core of the system is the Ethereum blockchain, where smart contracts written in Solidity automate critical functions such as voter registration, vote recording, and result tallying. Each vote is securely recorded on the blockchain, ensuring immutability, transparency, and preventing tampering. This architecture ensures secure, decentralized management of the voting process while providing a scalable and transparent solution.

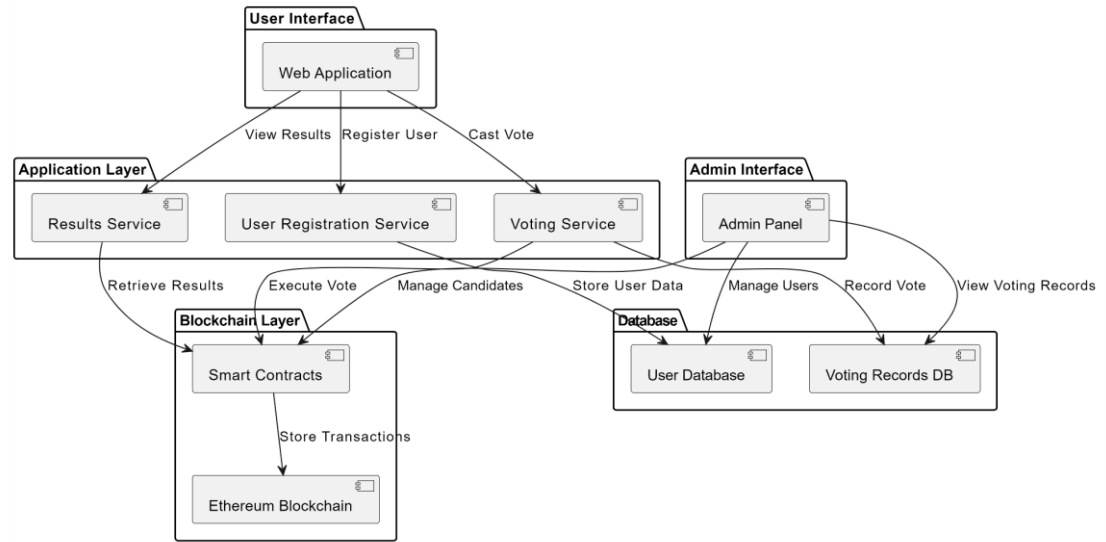


Fig 3.3.1: Proposed system Architecture

3.3.2 Working of the System

User Registration:

- Voters fill out a registration form with their details (e.g., Aadhar number).
- The system validates the information and records the voter on the blockchain.

User Login:

- Registered users log in by entering their Aadhar number and password.
- The system verifies credentials and grants access to the dashboard.

View Candidates:

- Users can view a list of candidates contesting in the election.
- The list is fetched from the blockchain for real-time accuracy.

Cast Vote:

- Users select a candidate and submit their vote.
- The system verifies that the user is eligible to vote (one vote per account).
- The vote is recorded on the blockchain through a smart contract.

Confirmation:

- After voting, the system sends a confirmation email to the voter.
- The vote is confirmed and stored immutably on the blockchain.

View Results:

- Users and admins can view the election results.
- The results are fetched directly from the blockchain, ensuring transparency.

Admin Functions:

- Admins log in to manage the system, add candidates, and view registered users.
- Any updates to candidates are also recorded on the blockchain.

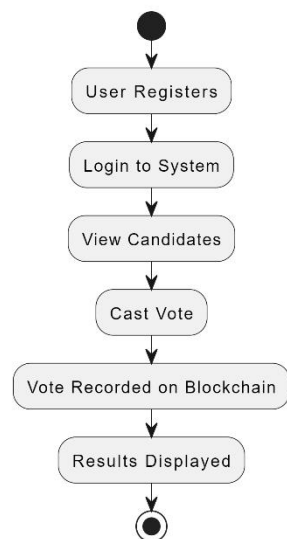


Fig 3.3.2: Process Flow

3.3.3 Flowchart of the system

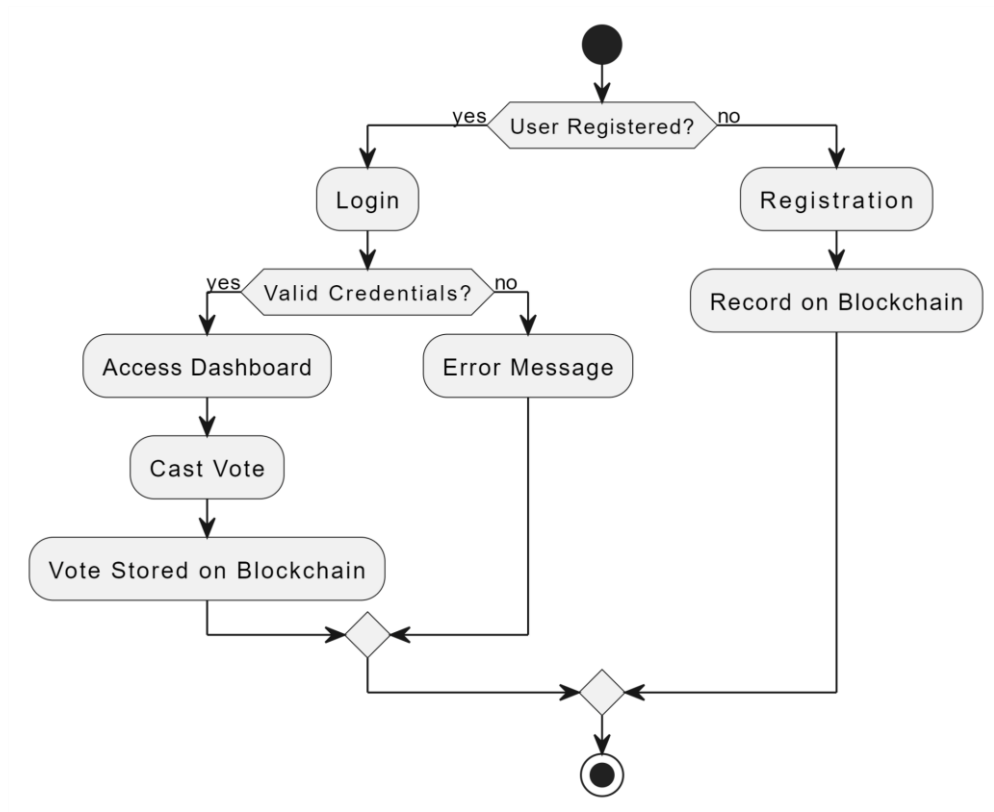


Fig 3.3.3 : Flowchart of system

Chapter IX

SYSTEM REQUIREMENTS AND SPECIFICATIONS

4.1 Software Requirements And Specifications

4.1.1 Introduction

Intended Audience and Reading Suggestions:

The intended audience for this document includes all stakeholders involved in the development and usage of the Online Voting System, specifically:

1. **Developers:** Responsible for implementing and maintaining the system.
2. **Users/Voters:** Individuals who will register and participate in the voting process.
3. **Administrators:** Responsible for managing users and overseeing the voting process.
4. **Testers:** Ensuring system functionality and performance through rigorous testing.
5. **Documentation Writers:** Creating user manuals and technical documentation.

All users are expected to have basic knowledge of computers and internet browsing. Administrators are assumed to have more advanced technical skills to troubleshoot issues and manage the system.

4.1.2 User Classes and Characteristics

There are three primary user classes in the system:

- **User (Voter):**
 - Access the voting platform via a web browser.
 - Register for voting.
 - Log in to the system using unique credentials.
 - View available candidates and cast votes.

- Receive confirmation of the vote via email.
- **Admin:**
 - Log in to the admin panel for managing the voting process.
 - Approve user registrations and verify identities.
 - Manage candidates and oversee election logistics.
 - Access and analyze voting results.
- **System:**
 - Process user registrations and login attempts.
 - Record votes securely on the blockchain.
 - Ensure integrity and transparency of the voting process.
 - Update the blockchain with election results.

4.1.3 Operating Environment

The system is a web-based application that can operate on any operating system with a modern web browser. Users must have an internet connection to access the voting platform and participate in elections.

4.1.4 External Interface Requirements

- **User Interfaces:** The user interface will consist of web pages for registration, login, and voting. Users will need to upload necessary documents (for identity verification) and complete online voting.
- **Hardware Interfaces:**
 - Client PC or mobile device with an internet connection.
- **Software Requirements:**
 - Development Tools: Node.js, Express.js, MongoDB, Solidity, Truffle, Visual Studio Code, Postman, and any modern web browser.

4.1.5 Functional Requirements

- **User Registration:** Users must complete a registration form, which should be validated and saved in the blockchain.

- **User Login:** Users must enter valid credentials to access their accounts. Incorrect credentials will prevent access.
- **Voting:** Users can view candidates and cast their votes. The system ensures only one vote per registered user.
- **Result Viewing:** Users can access election results displayed in real-time, directly fetched from the blockchain.

4.1.6 Other Non-functional Requirements

- **Data Input Requirements:**
 - Users must upload scanned documents for identity verification, adhering to specified size and format.
- **Security Requirements:**
 - The system must ensure authentication for authorized users only.
 - Protection against security attacks (e.g., DDoS, SQL injection).
 - Admin passwords and sensitive user information must be securely encrypted.
- **Software Quality Attributes:**
 - Reliability: The system should operate consistently without failure.
 - Efficiency: Quick response times for user actions.
 - Security: Protection of user data and votes.
 - User-friendliness: Intuitive interface for all users.
 - Flexibility: Adaptability to various election types and scales.

4.1.7 Product Perspective

The Online Voting System is a web-based application implementing a client-server model that utilizes blockchain technology to ensure a secure and transparent voting process. The system supports:

- **User Accounts:** Users can create accounts, update profiles, and track their voting history.

- **Scalability:** The system can accommodate a large number of simultaneous users during elections.
- **Search Functionality:** Users can search for candidates based on keywords or criteria.
- **Feedback Mechanism:** A platform for users to provide feedback on candidates and the voting process.
- **FAQs Section:** Answers to common questions about the voting process and system functionality.

4.1.8 Product Function

Key functionalities of the Online Voting System include:

- **Voting Details:**
 - Candidate information, voting status, and unique voting ID.
- **User Information:**
 - User ID, registration details, and vote history.
- **Election Overview:**
 - Election ID, date, total registered voters, and results summary.

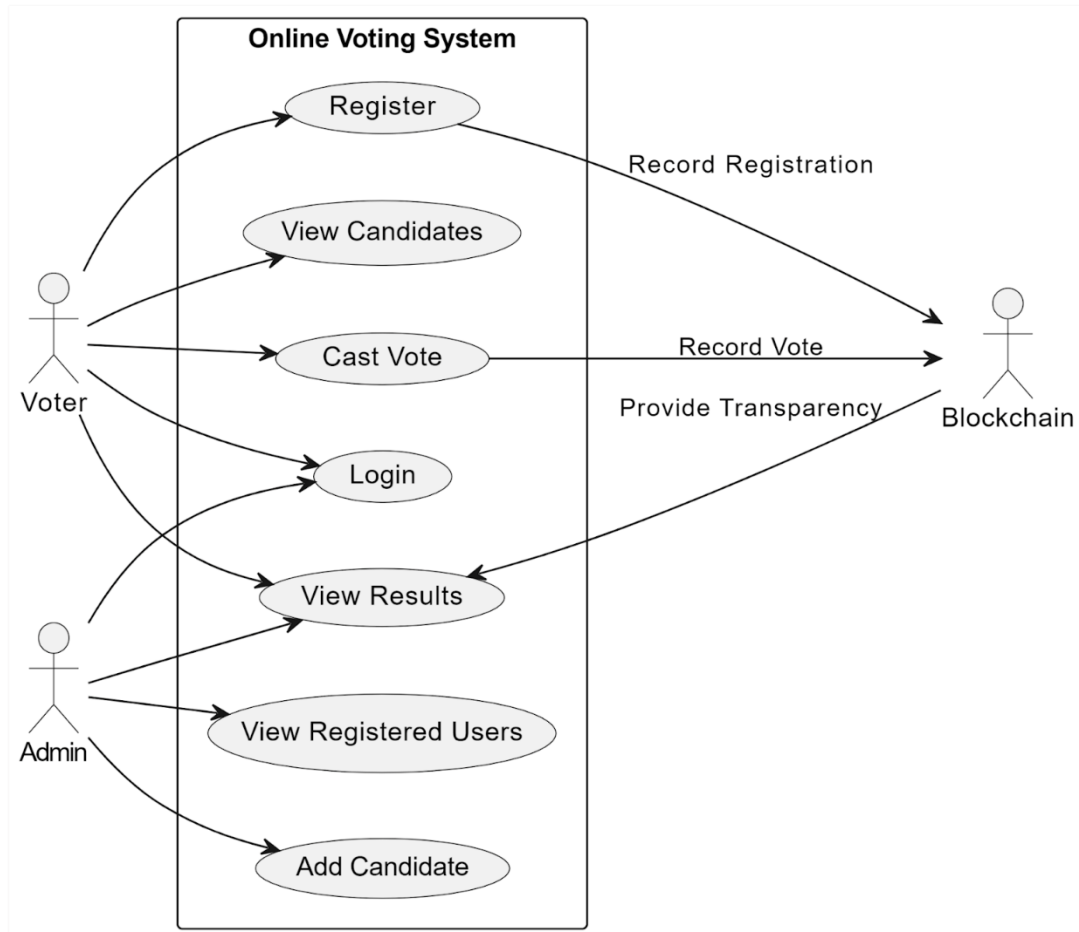
4.1.9 Modules – Developed for Business Logic

Module Name	Description
User Management Module	Handles user registration, login, and profile management. Validates user information and maintains records.
Voting Module	Manages the entire voting process, including displaying candidates, casting votes, and ensuring one vote per user. It records votes securely on the blockchain.
Admin Management Module	Facilitates admin tasks such as user approval, candidate management, and overseeing the election process. Allows admins to access voting results.
Blockchain Integration Module	Interfaces with the Ethereum blockchain to handle smart contracts for secure vote recording and retrieval. Ensures data integrity and transparency.
Reporting Module	Generates comprehensive reports on voter statistics, election results, and system performance metrics for analysis and review.

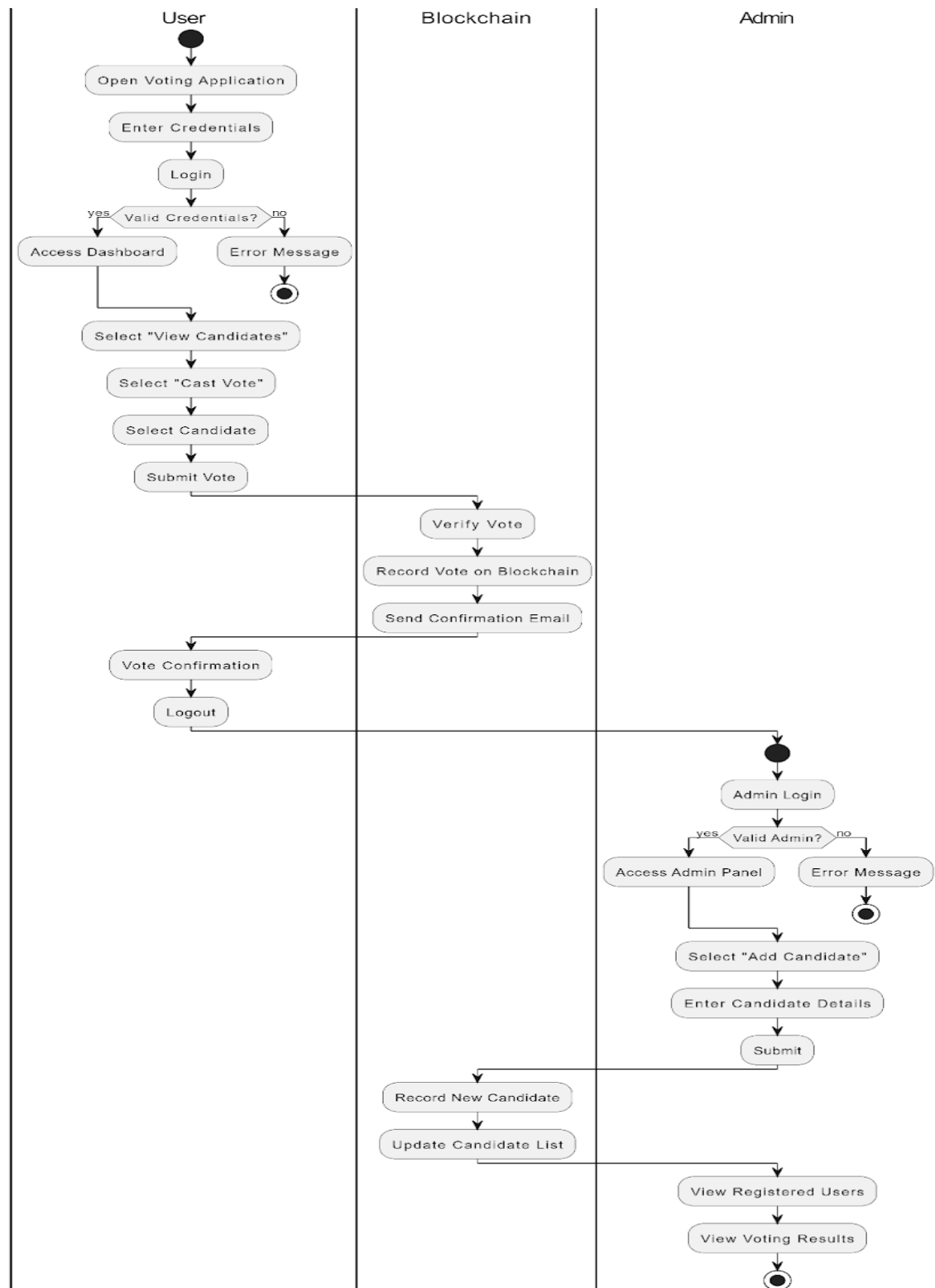
Chapter X

PROJECT ANALYSIS AND DESIGN

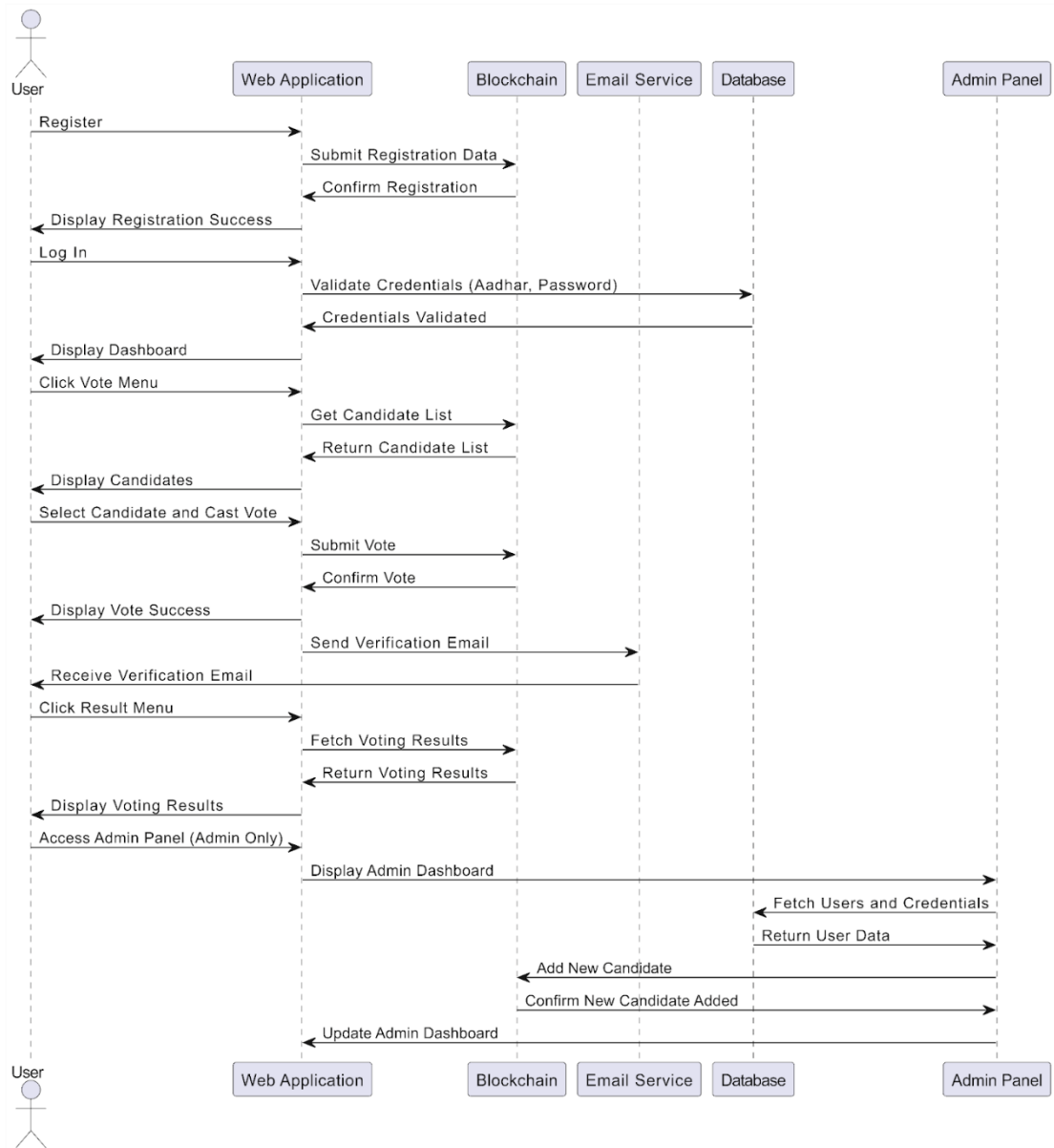
5.1 USE CASE DIAGRAM



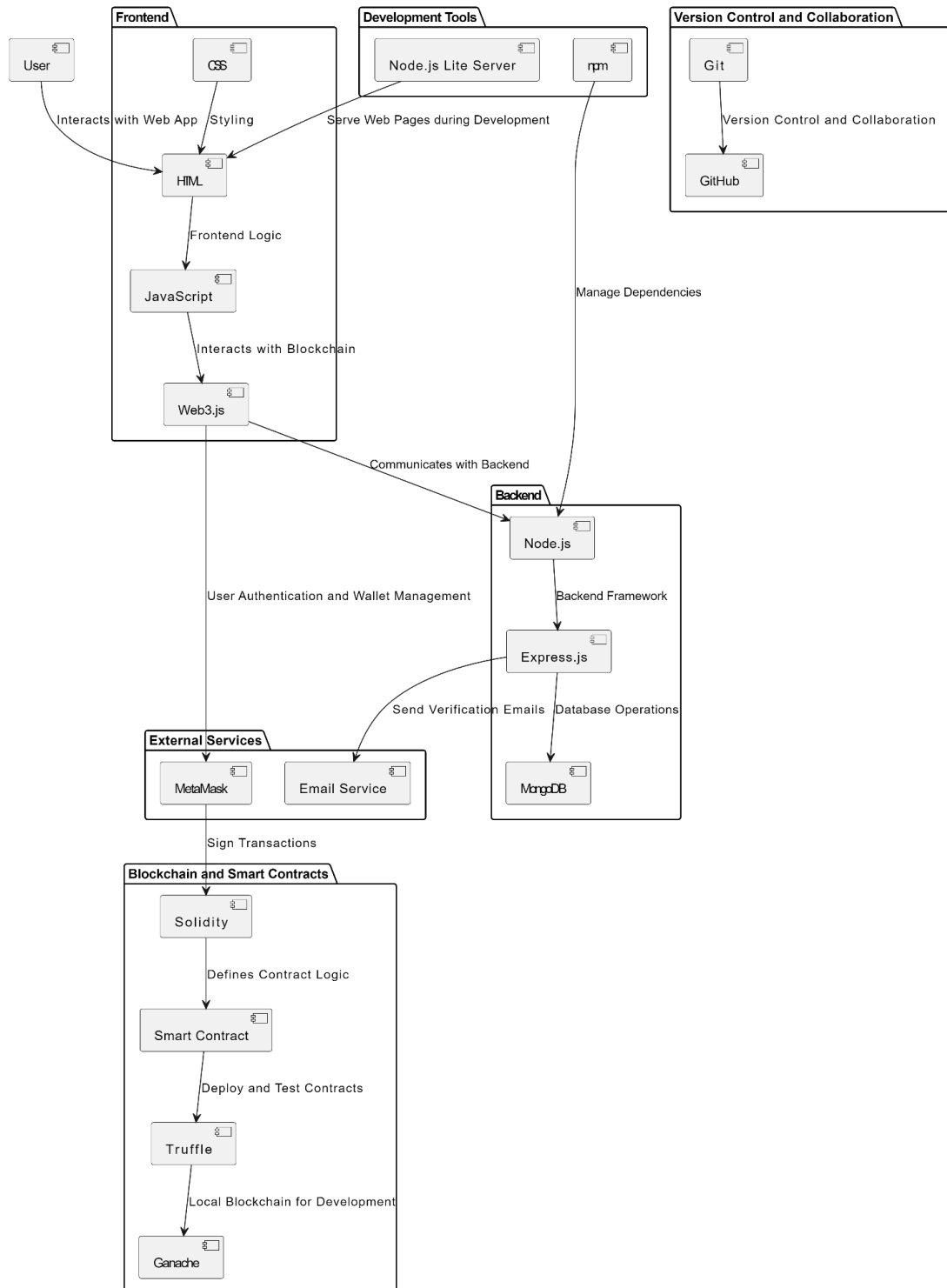
5.2 ACTIVITY DIAGRAM



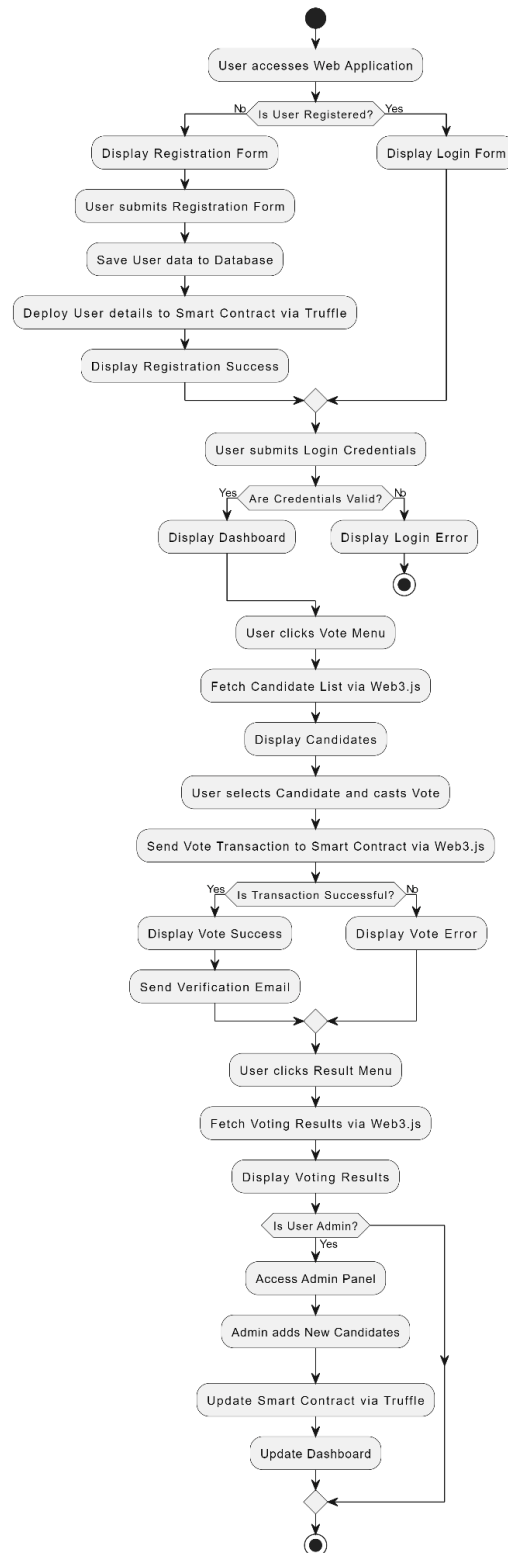
5.3 SEQUENCE DIAGRAM



5.4 WORKFLOW



5.5 Flow Diagram



5.6 TIME SCHEDULE

Task	Start Date	End Date	Status
Project Initiation	June 15	June 20	Completed
Requirement Gathering	June 21	June 30	Completed
System Design	July 1	July 10	Completed
Development Phase 1	July 11	July 20	Completed
Development Phase 2	July 21	July 31	Completed
Testing Phase 1	August 1	August 7	Completed
Testing Phase 2	August 8	August 15	Completed
Deployment	August 16	August 20	Completed
User Training	August 21	August 25	Completed
Project Closure	August 26	August 30	Completed

5.6 TEAM ORGANISATION

5.6.1 Team Structure

Our team consists of developers, internal guide, external guide and One mentor from Vijay Computers and data systems.

Developers:

Mr. Tanish Bagal

Mr. Harshal Dahiwale

Mr. Shrinivas Kathare

Mr. Shreyas Raijade

Internal Guide: Prof. Shalini V. Wankhede

External Guide:

Vijay Computers and data systems mentor: Ms. Vaibhavi Kadam

5.6.2 Agile Methodology Practice in the Online Voting System Project

The Agile methodology was employed in the development of the Online Voting System to ensure flexibility, collaboration, and iterative progress. Here's a brief overview of how Agile practices were applied:

1. Sprint Planning:

- The project was divided into multiple sprints, each lasting 2-4 weeks. During each sprint planning session, specific features related to user management, voting processes, and admin functionalities were prioritized based on stakeholder feedback.

2. Daily Stand-Ups:

- Short daily meetings were held to discuss progress, address challenges, and ensure alignment among team members. This practice encouraged quick resolution of issues and fostered collaboration.

3. User Stories:

- User stories were created to capture requirements from the perspective of different users (voters, admins). Each story outlined the desired functionality, ensuring that development focused on user needs and expectations.

4. Iterative Development:

- Features were developed in iterations, allowing for continuous testing and feedback. After each sprint, a review meeting was conducted to demonstrate completed features to stakeholders and gather input for the next iteration.

5. Continuous Integration and Testing:

- Code changes were frequently integrated into a shared repository, enabling automated testing to identify bugs early in the development process. This ensured a stable and reliable system throughout development.

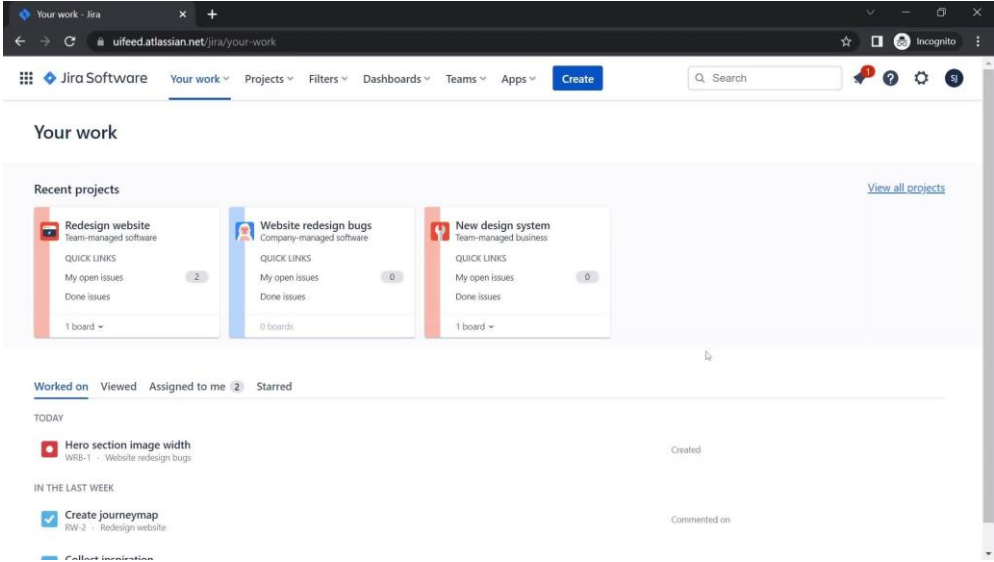
6. Feedback Loops:

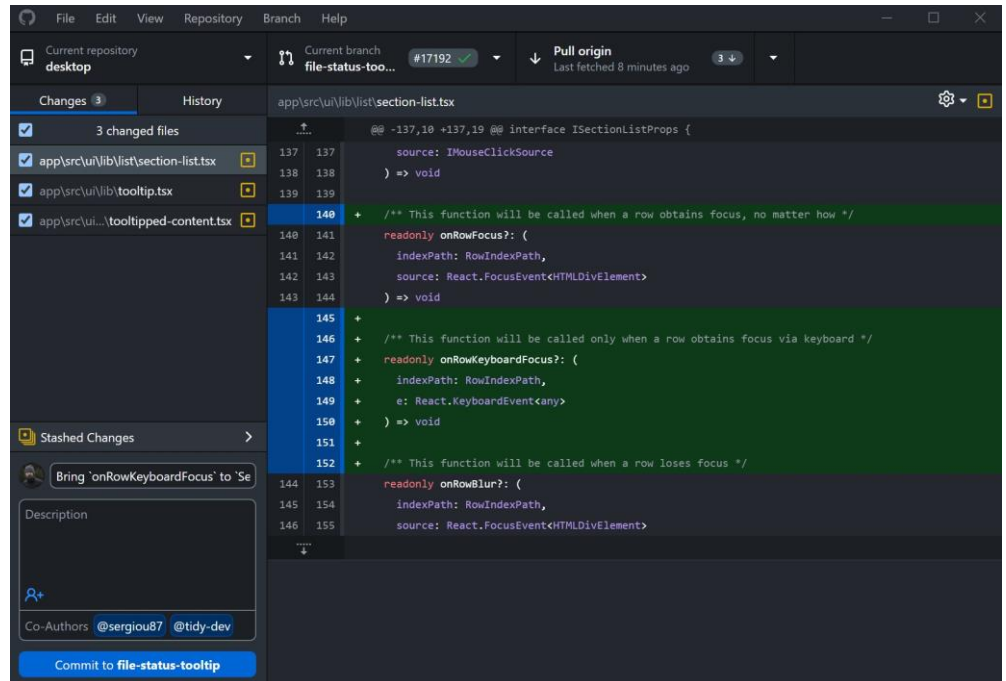
- Regular feedback from users and stakeholders was incorporated after each sprint, allowing the team to adapt to changing requirements and improve the system iteratively.

7. Retrospectives:

- At the end of each sprint, the team conducted retrospective meetings to reflect on what worked well and what could be improved. This practice promoted a culture of continuous improvement within the team.

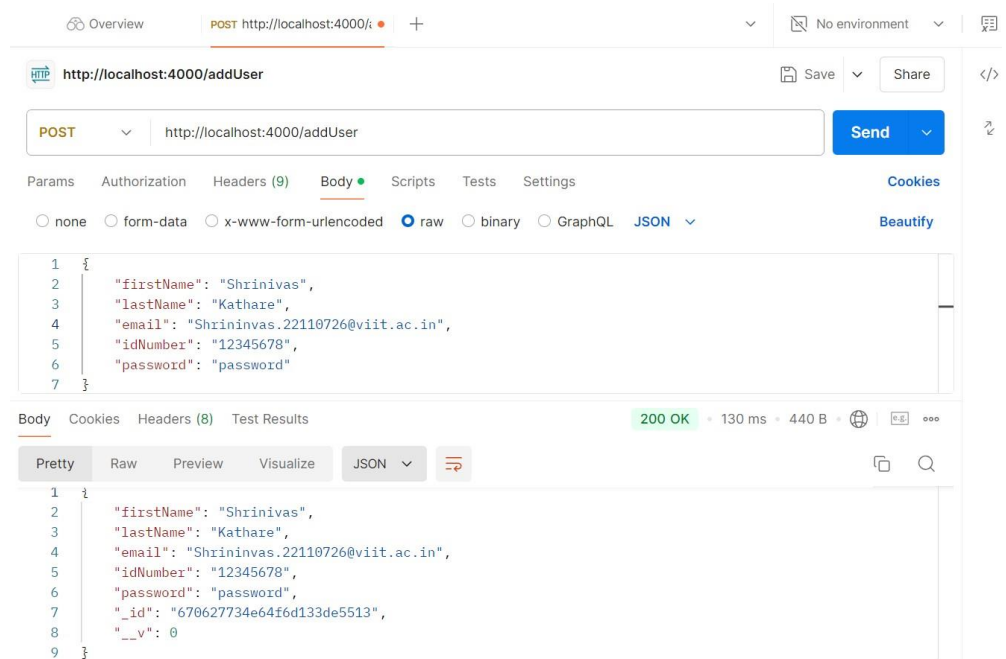
5.6.3

Tool	Description
Jira	A project management tool that helps in planning, tracking, and managing agile software development projects. It is used for issue tracking, task assignments, and sprint planning. 
Google Meet	A collaboration platform that facilitates communication and teamwork among project members through chat, video conferencing, and file sharing. It serves as a hub for discussions and meetings.
GitHub	A version control platform that enables code collaboration, version tracking, and repository management. It allows developers to manage code changes, track issues, and collaborate on code reviews.



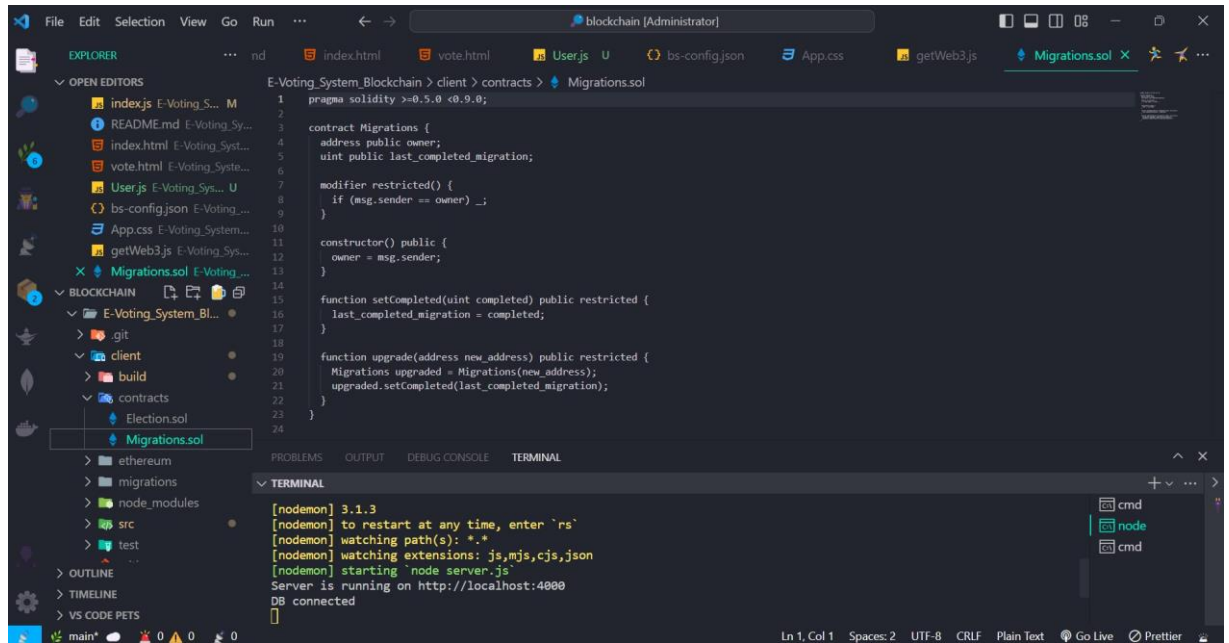
Postman

An API development tool that simplifies the process of testing and documenting APIs. It is used for sending requests to the backend services and validating responses, ensuring the backend functionalities work as expected.



Visual Studio Code

A lightweight and powerful code editor that supports multiple programming languages. It is utilized for writing, debugging, and managing code, providing features like extensions, syntax highlighting, and integrated terminal.



The screenshot displays a Visual Studio Code editor window with a project named 'E-Voting_System_Blockchain'. The Explorer sidebar on the left shows the project structure, including folders for 'client', 'build', 'contracts', and 'ethereum', as well as files like 'index.html', 'vote.html', 'User.js', 'bs-config.json', 'App.css', 'getWeb3.js', and 'Migrations.sol'. The main editor area shows the 'Migrations.sol' file with the following Solidity code:

```
1 pragma solidity >=0.5.0 <0.9.0;
2
3 contract Migrations {
4     address public owner;
5     uint public last_completed_migration;
6
7     modifier restricted() {
8         if (msg.sender == owner) _;
9     }
10
11     constructor() public {
12         owner = msg.sender;
13     }
14
15     function setCompleted(uint completed) public restricted {
16         last_completed_migration = completed;
17     }
18
19     function upgrade(address new_address) public restricted {
20         Migrations upgraded = Migrations(new_address);
21         upgraded.setCompleted(last_completed_migration);
22     }
23 }
24
```

The bottom panel shows the 'TERMINAL' window with the following output:

```
[nodemon] 3.1.3
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
Server is running on http://localhost:4000
DB connected
```

5.6.4 Learning Sessions

1. Blockchain Basics:
 - Explored blockchain security, transparency, and Ethereum smart contracts.
2. Smart Contracts:
 - Learned Solidity for developing and deploying voting smart contracts.
3. Web Development:
 - Enhanced skills in HTML, CSS, and JavaScript for front-end design.
 - Used Node.js and Express.js for back-end development.
4. Database Management:
 - Gained experience with MongoDB for user data storage.
5. Security Practices:
 - Implemented user authentication and encryption techniques.
6. Testing Skills:
 - Conducted unit and integration testing using tools like Postman.
7. Project Management:
 - Utilized Jira for task management and agile practices.
8. User Experience:
 - Focused on UX design and gathered user feedback for improvements.

5.7 INTERFACE DETAILS AND SCREENSHOTS

The screenshot shows the 'Create an account' registration form on the BLOCKCHAIN-VOTE website. The form is set against a dark blue background with a network pattern. At the top, the website name 'BLOCKCHAIN-VOTE' is on the left, and navigation links 'Register', 'Login', 'About', 'Results', and 'Admin Panel' are on the right. The main heading 'Create an account' is centered. The form fields include: 'First Name' (Shrinivas), 'Last Name' (Kathare), 'Aadhar Number' (412897271457), 'Mobile No.' (9511860406), 'Email' (shrinivas.22110726@viit.ac.in), and 'Password' (masked with dots). A red 'Register' button is below the password field. Below the button, a link says 'Already have an account? Click here to login'. At the bottom, a red box displays 'Your Account: 0xad1a20f668ag8ccecff10aabf3aec07c808a226'.

Fig 5.7.1 Sign-in

The screenshot shows the 'Login to an existing account' form on the BLOCKCHAIN-VOTE website. The layout is similar to the registration page. The main heading 'Login to an existing account' is centered. The form fields include: 'Aadhar Number' (shrinivas.22110726@viit.ac.in) and 'Password' (masked with dots). A red 'Login' button is below the password field. At the bottom, a red box displays 'Your Account: 0xad1a20f668ag8ccecff10aabf3aec07c808a226'.

Fig 5.7.2 Login

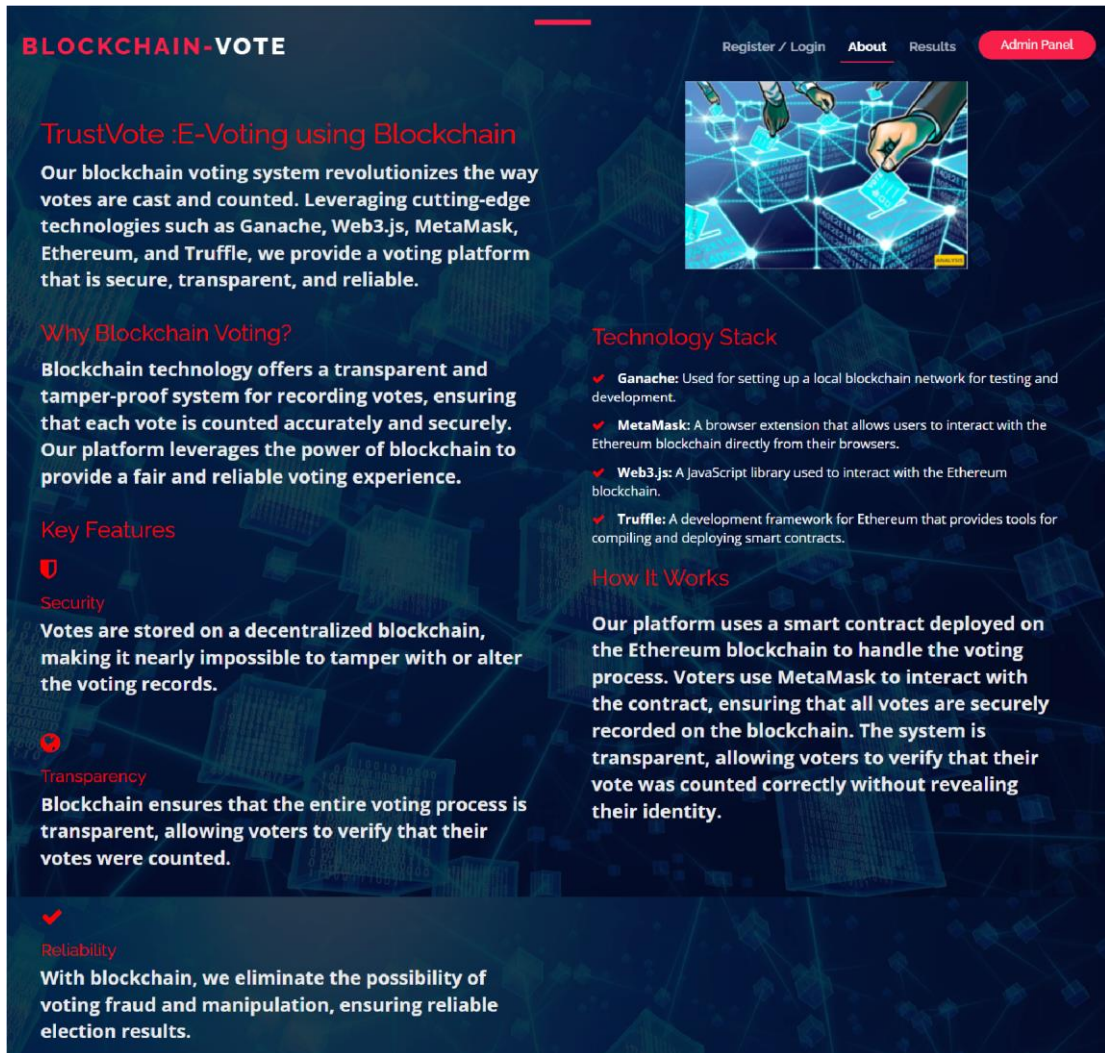


Fig 5.6.3 About Page

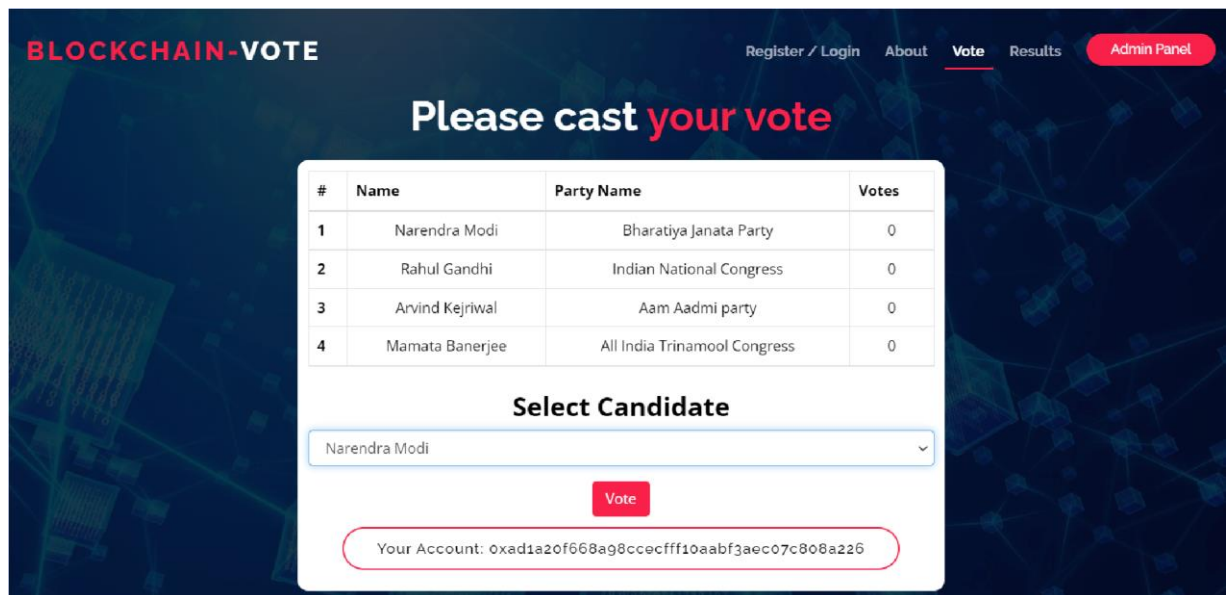


Fig 5.6.4 Vote Page

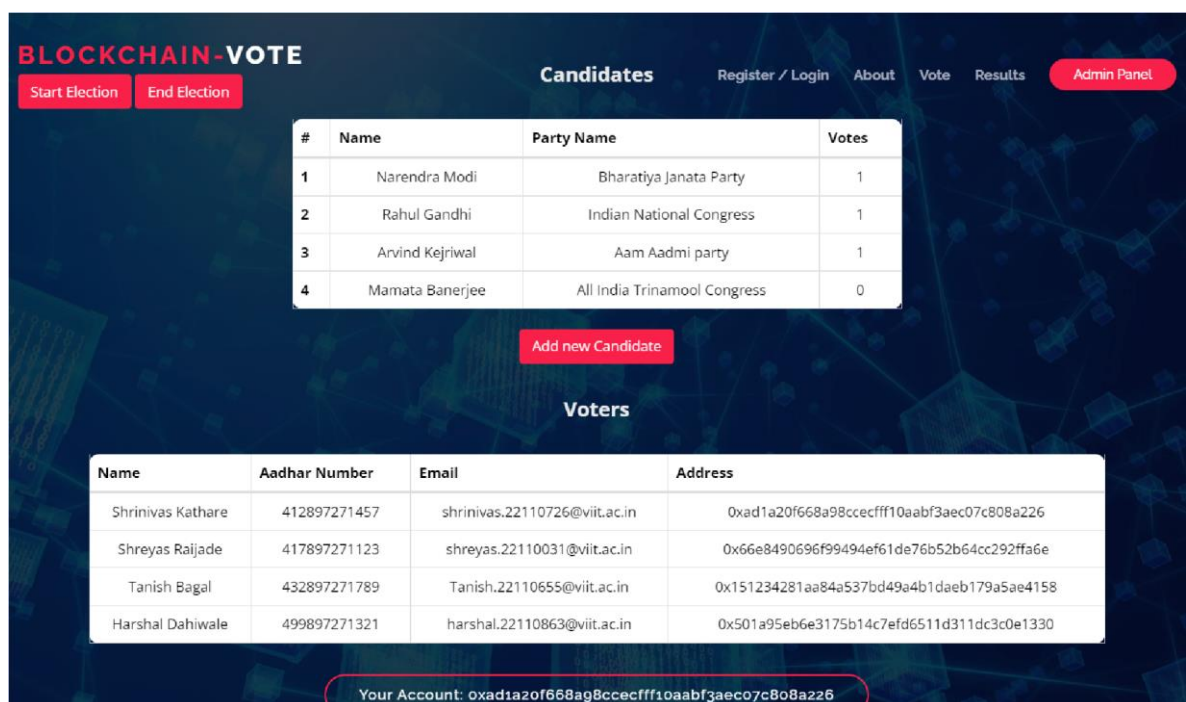


Fig 5.6.5 Admin Panel

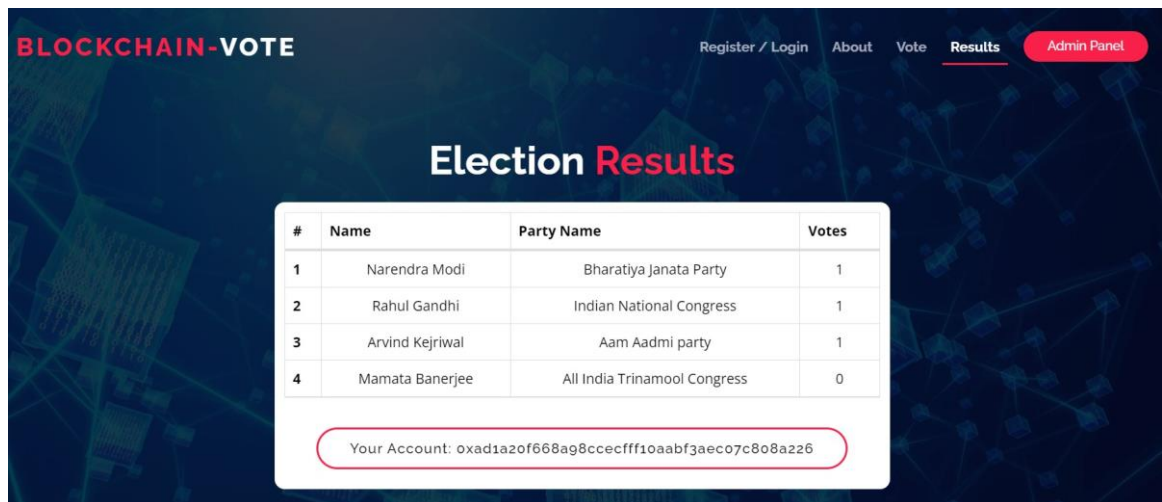
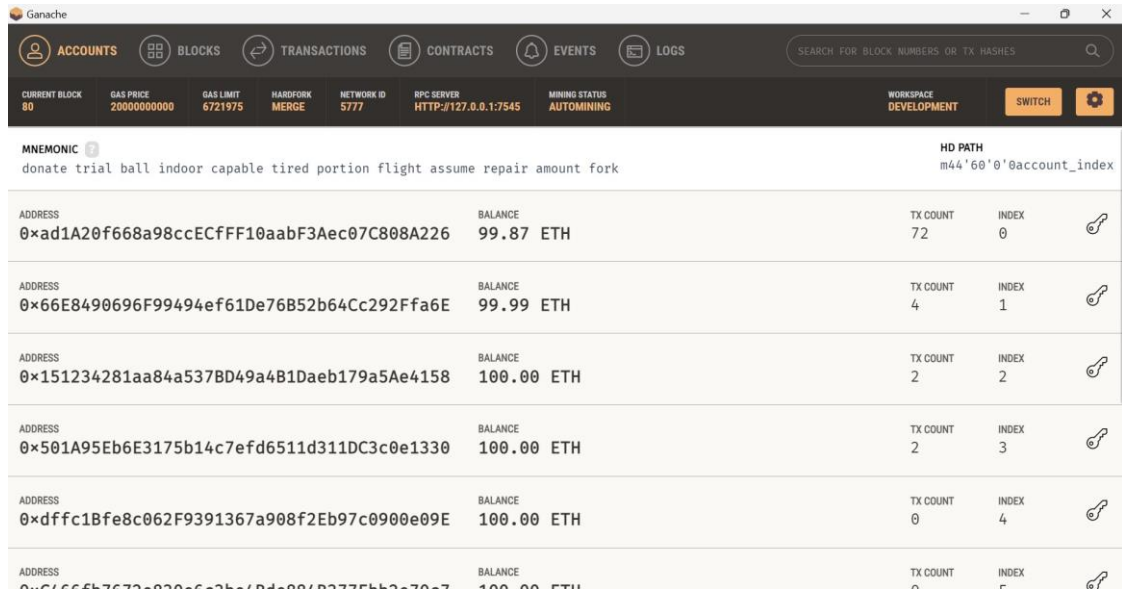


Fig 5.6.6 Result Page

Chapter XI

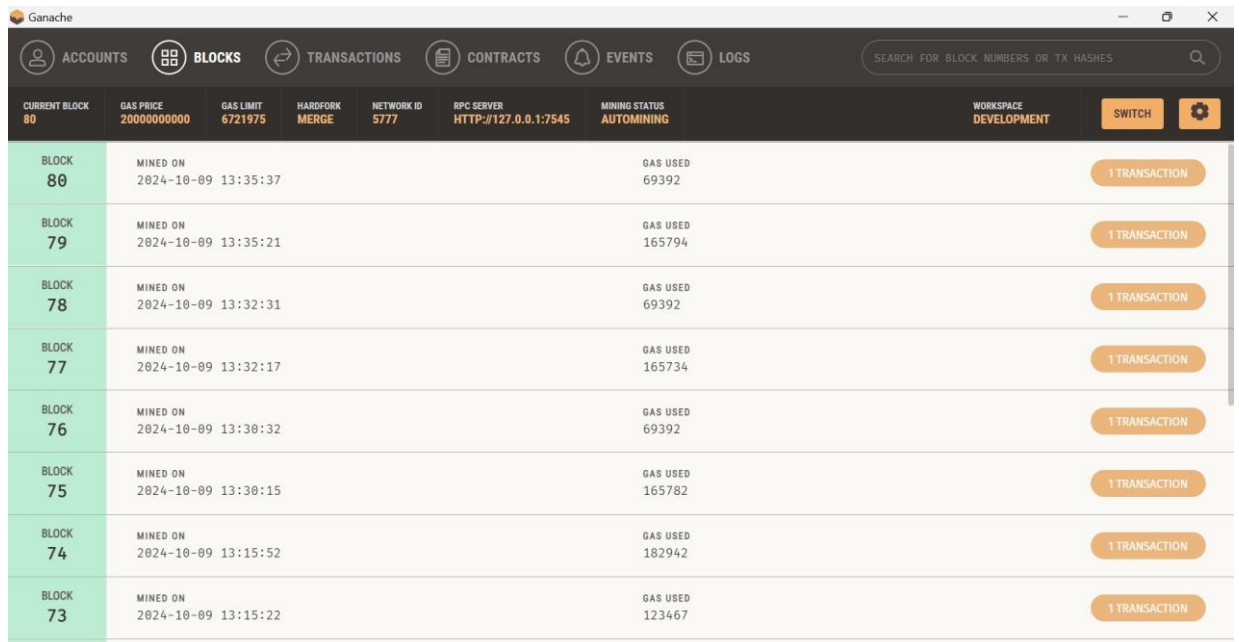
IMPLEMENTATION SOFTWARE TESTING

6.1 IMPLEMENTATION



ADDRESS	BALANCE	TX COUNT	INDEX
0xad1A20f668a98ccECfFF10aabF3Aec07C808A226	99.87 ETH	72	0
0x66E8490696F99494ef61De76B52b64Cc292Ffa6E	99.99 ETH	4	1
0x151234281aa84a537BD49a4B1Daeb179a5Ae4158	100.00 ETH	2	2
0x501A95Eb6E3175b14c7efd6511d311DC3c0e1330	100.00 ETH	2	3
0xdffc1Bfe8c062F9391367a908f2Eb97c0900e09E	100.00 ETH	0	4
0x151234281aa84a537BD49a4B1Daeb179a5Ae4158	100.00 ETH	2	2

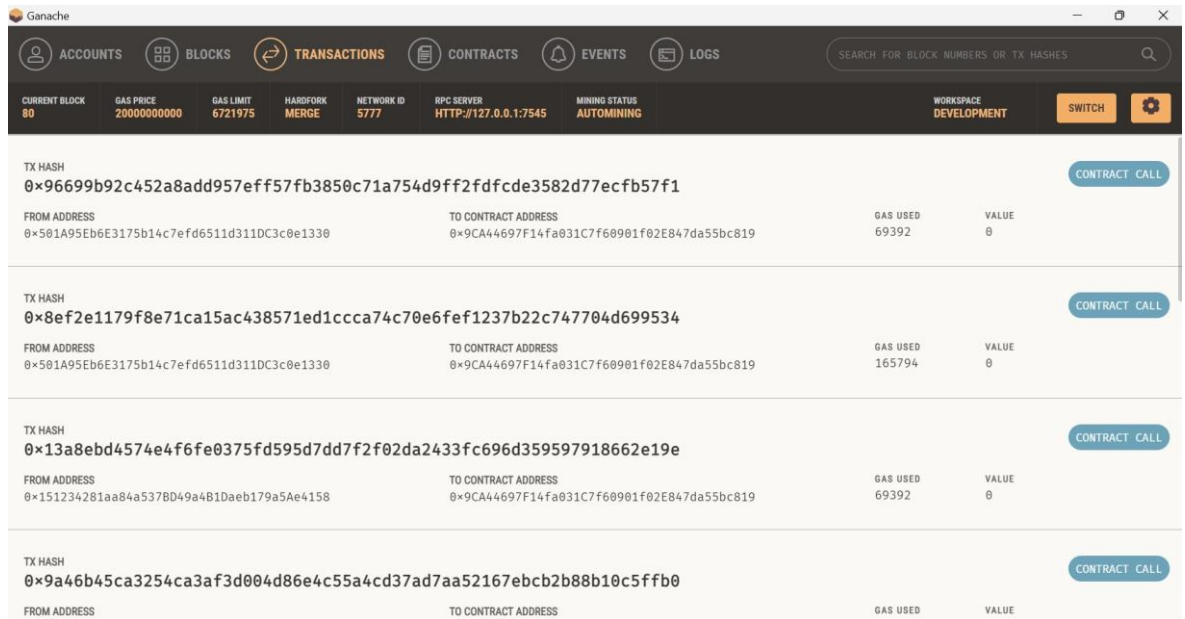
Fig.6.1.1 Ganache



BLOCK	MINED ON	GAS USED	TRANSACTIONS
80	2024-10-09 13:35:37	69392	1 TRANSACTION
79	2024-10-09 13:35:21	165794	1 TRANSACTION
78	2024-10-09 13:32:31	69392	1 TRANSACTION
77	2024-10-09 13:32:17	165734	1 TRANSACTION
76	2024-10-09 13:30:32	69392	1 TRANSACTION
75	2024-10-09 13:30:15	165782	1 TRANSACTION
74	2024-10-09 13:15:52	182942	1 TRANSACTION
73	2024-10-09 13:15:22	123467	1 TRANSACTION

Fig 6.1.2 Blocks of blockchain

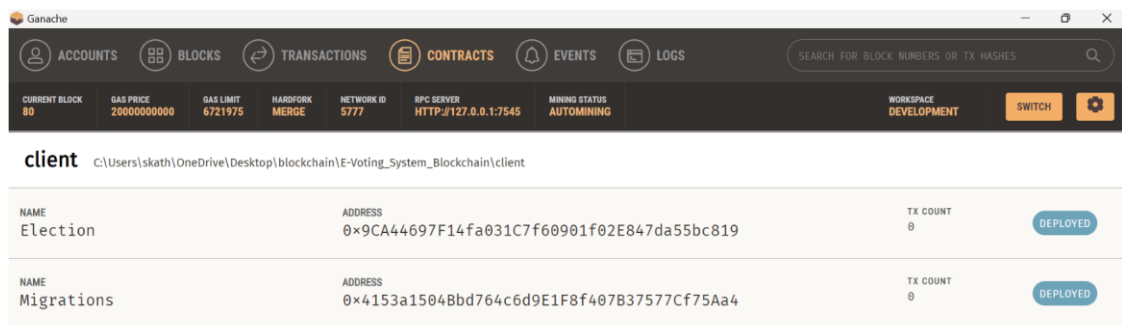
TrustVote: Online Voting Using BlockChain



The screenshot shows the Ganache application window with the 'TRANSACTIONS' tab selected. The top bar displays various network statistics: Current Block 80, Gas Price 20000000000, Gas Limit 6721975, Hardfork MERGE, Network ID 5777, RPC Server HTTP://127.0.0.1:7545, Mining Status AUTOMINING, and Workspace DEVELOPMENT. Below the top bar, there are four transaction entries, each with a 'CONTRACT CALL' button. The transactions are as follows:

TX HASH	FROM ADDRESS	TO CONTRACT ADDRESS	GAS USED	VALUE
0x96699b92c452a8add957eff57fb3850c71a754d9ff2dfdcde3582d77ecfb57f1	0x501A95Eb6E3175b14c7ef6d511d311DC3c0e1330	0x9CA44697F14fa031C7f60901f02E847da55bc819	69392	0
0x8ef2e1179f8e71ca15ac438571ed1ccca74c70e6fef1237b22c747704d699534	0x501A95Eb6E3175b14c7ef6d511d311DC3c0e1330	0x9CA44697F14fa031C7f60901f02E847da55bc819	165794	0
0x13a8ebd4574e4f6fe0375fd595d7dd7f2f02da2433fc696d359597918662e19e	0x151234281aa84a537BD49a4B1Daeb179a5Ae4158	0x9CA44697F14fa031C7f60901f02E847da55bc819	69392	0
0x9a46b45ca3254ca3af3d004d86e4c55a4cd37ad7aa52167ebcb2b88b10c5fffb0				

Fig 6.1.3 Blockchain Transactions

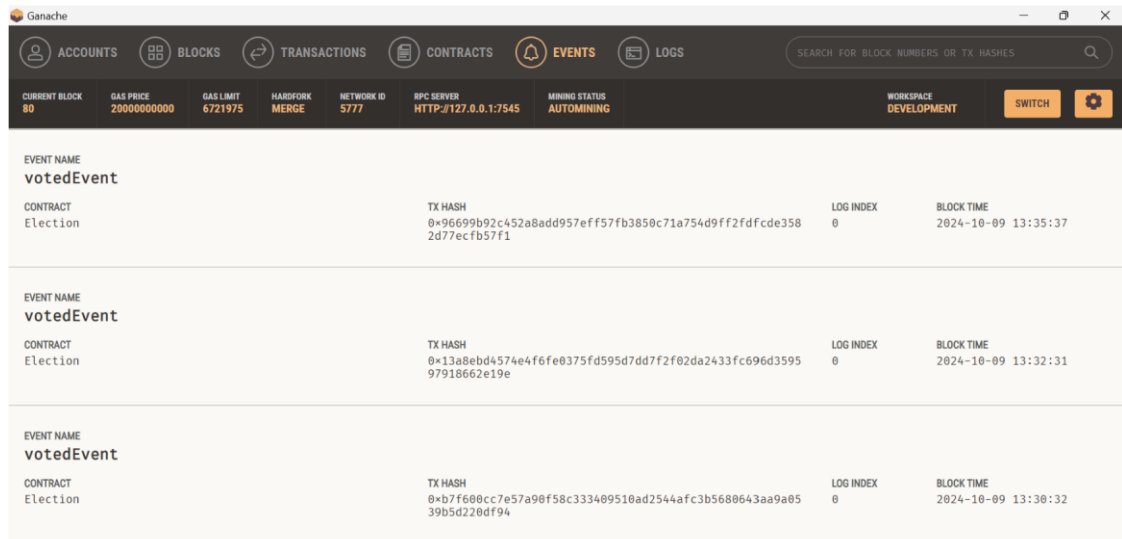


The screenshot shows the Ganache application window with the 'CONTRACTS' tab selected. The top bar displays the same network statistics as the previous screenshot. Below the top bar, there are two contract entries, each with a 'DEPLOYED' button. The contracts are as follows:

NAME	ADDRESS	TX COUNT
Election	0x9CA44697F14fa031C7f60901f02E847da55bc819	0
Migrations	0x4153a1504Bbd764c6d9E1F8f407B37577Cf75Aa4	0

Fig 6.1.4 Blockchain Contracts

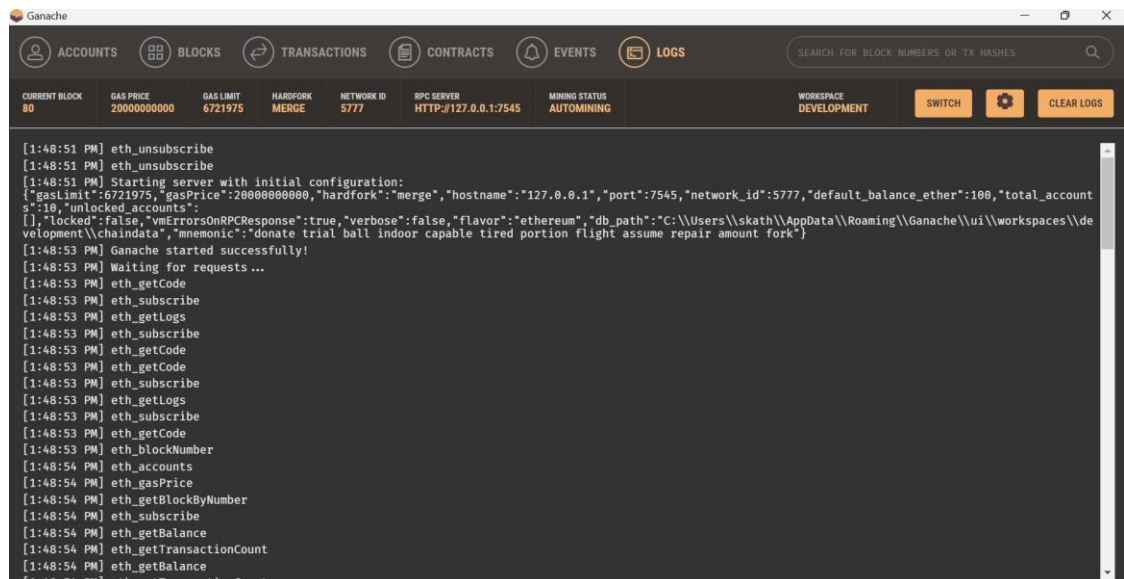
TrustVote: Online Voting Using BlockChain



The screenshot shows the Ganache application window. The top navigation bar includes icons for ACCOUNTS, BLOCKS, TRANSACTIONS, CONTRACTS, EVENTS (highlighted), and LOGS. Below the navigation bar, a status bar displays various network metrics: CURRENT BLOCK 80, GAS PRICE 20000000000, GAS LIMIT 6721975, HARDFORK MERGE, NETWORK ID 5777, RPC SERVER HTTP://127.0.0.1:7545, MINING STATUS AUTOMINING, and WORKSPACE DEVELOPMENT. The main content area displays a list of events for the 'votedEvent' contract. Each event entry includes the event name, contract name, transaction hash, log index, and block time.

EVENT NAME	CONTRACT	TX HASH	LOG INDEX	BLOCK TIME
votedEvent	Election	0x96699b92c452a8add957eff57fb3850c71a754d9ff2fdcd3582d77ecfb57f1	0	2024-10-09 13:35:37
votedEvent	Election	0x13a8ebd4574e4f6fe0375fd595d7dd7f2f02da2433fc696d359597918662e19e	0	2024-10-09 13:32:31
votedEvent	Election	0xb7f600cc7e57a90f58c333409510ad2544afc3b5680643aa9a0539b5d228df94	0	2024-10-09 13:30:32

Fig 6.1.5 Blockchain Events



The screenshot shows the Ganache application window with the LOGS tab selected. The top navigation bar and status bar are identical to the previous figure. The main content area displays a list of log entries, each starting with a timestamp and followed by a log message. The logs show the initialization of the Ganache server, including configuration details and the start of the mining process.

Timestamp	Log Message
[1:48:51 PM]	eth_unsubscribe
[1:48:51 PM]	eth_unsubscribe
[1:48:51 PM]	Starting server with initial configuration: {"gasLimit":6721975,"gasPrice":20000000000,"hardfork":"merge","hostname":"127.0.0.1","port":7545,"network_id":5777,"default_balance_ether":100,"total_account":10,"unlocked_accounts":{}}
[1:48:53 PM]	Ganache started successfully!
[1:48:53 PM]	Waiting for requests ...
[1:48:53 PM]	eth_getCode
[1:48:53 PM]	eth_subscribe
[1:48:53 PM]	eth_getLogs
[1:48:53 PM]	eth_subscribe
[1:48:53 PM]	eth_getCode
[1:48:53 PM]	eth_getCode
[1:48:53 PM]	eth_subscribe
[1:48:53 PM]	eth_getLogs
[1:48:53 PM]	eth_subscribe
[1:48:53 PM]	eth_getCode
[1:48:53 PM]	eth_blockNumber
[1:48:54 PM]	eth_accounts
[1:48:54 PM]	eth_gasPrice
[1:48:54 PM]	eth_getBlockByNumber
[1:48:54 PM]	eth_subscribe
[1:48:54 PM]	eth_getBalance
[1:48:54 PM]	eth_getTransactionCount
[1:48:54 PM]	eth_getBalance

Fig 6.1.5 Blockchain logs

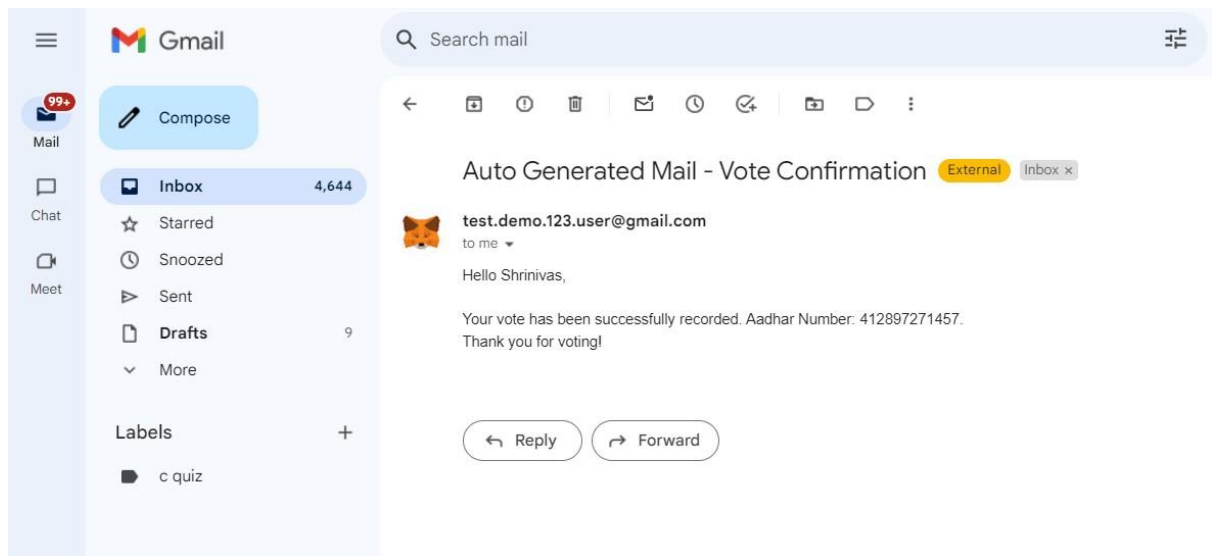


Fig 6.1.5 Vote Confirmation Mail

6.2 INTEGRATION

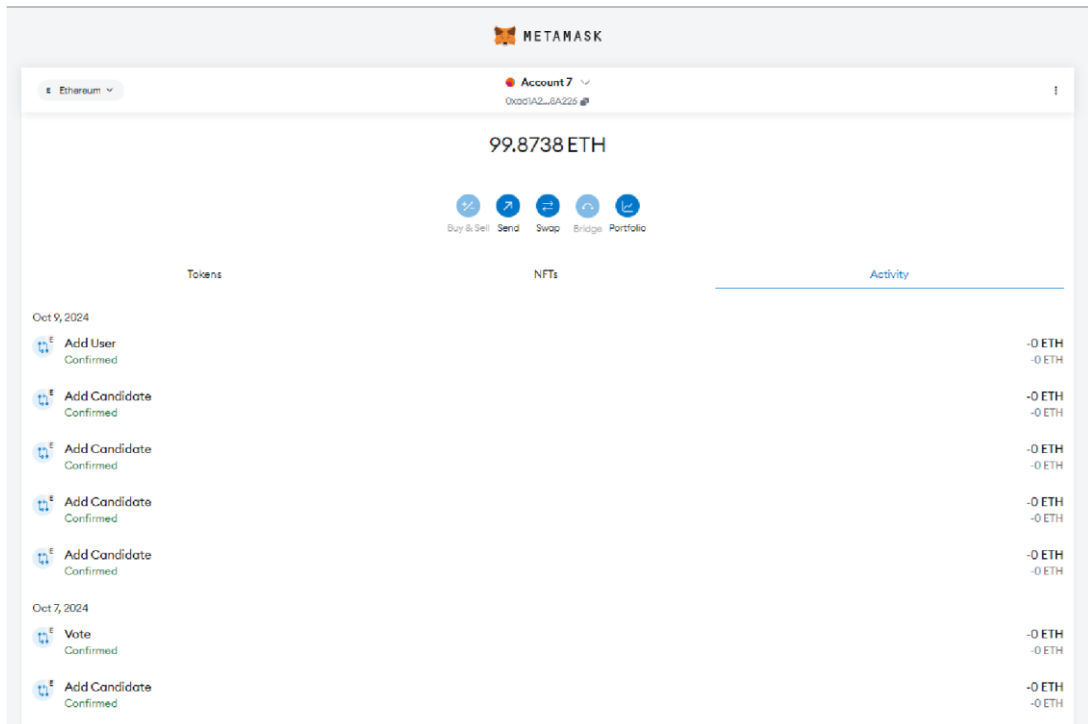


Fig 6.3.1 Metamask Integration

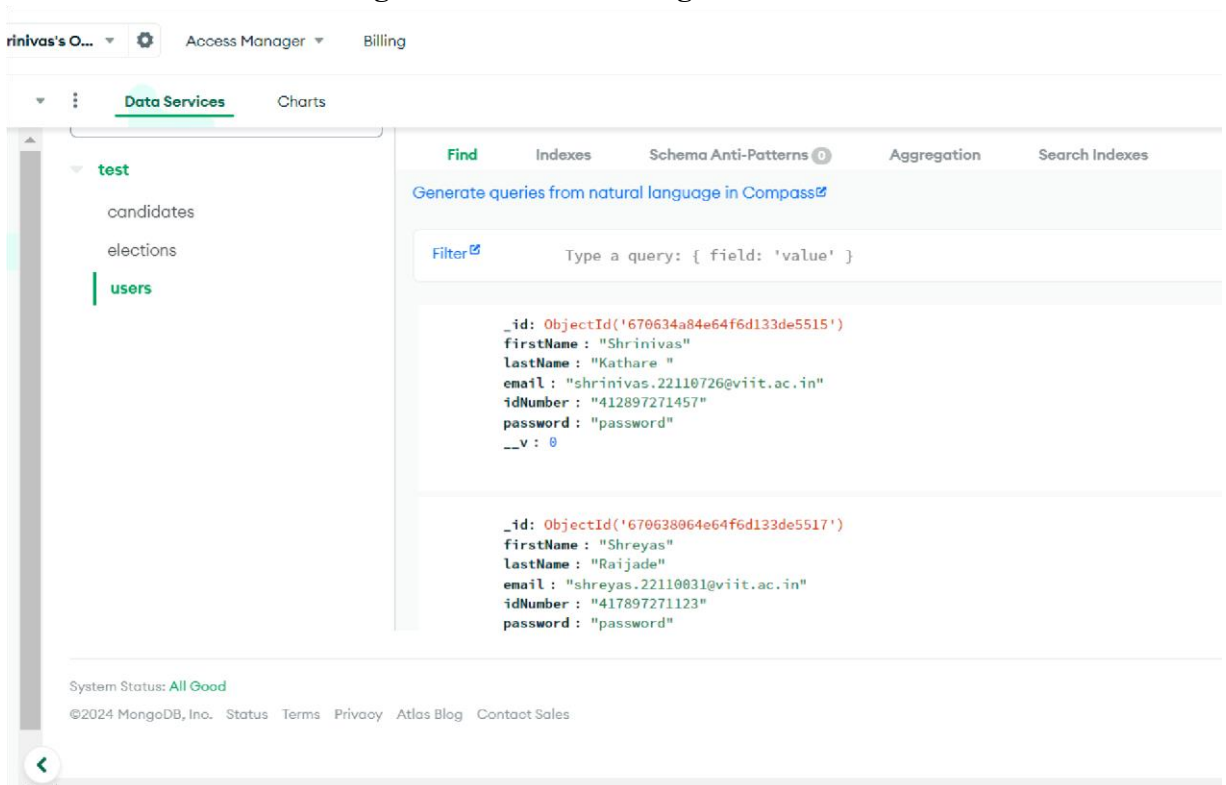


Fig 6.2. MongoDB Database

TrustVote: Online Voting Using BlockChain

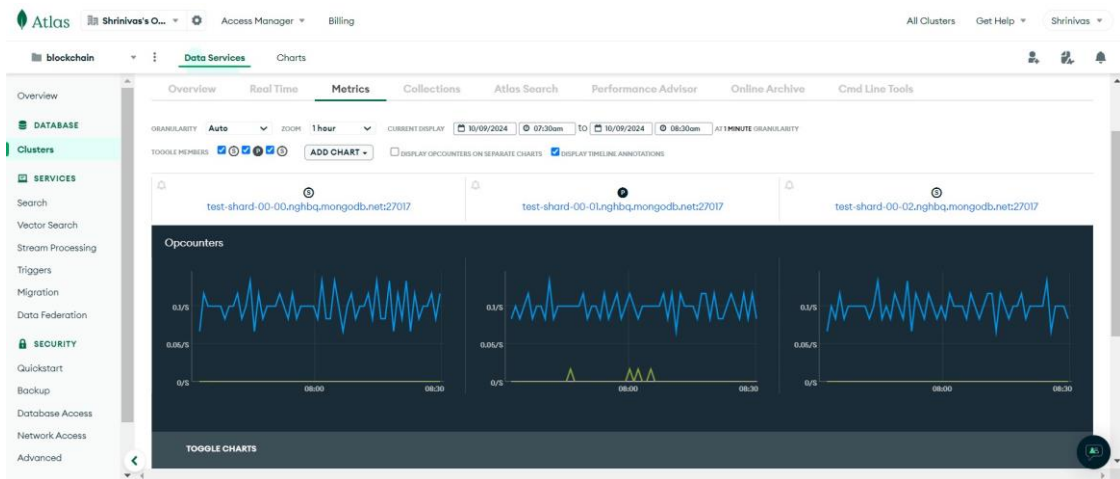


Fig 6.3. Database Monitoring

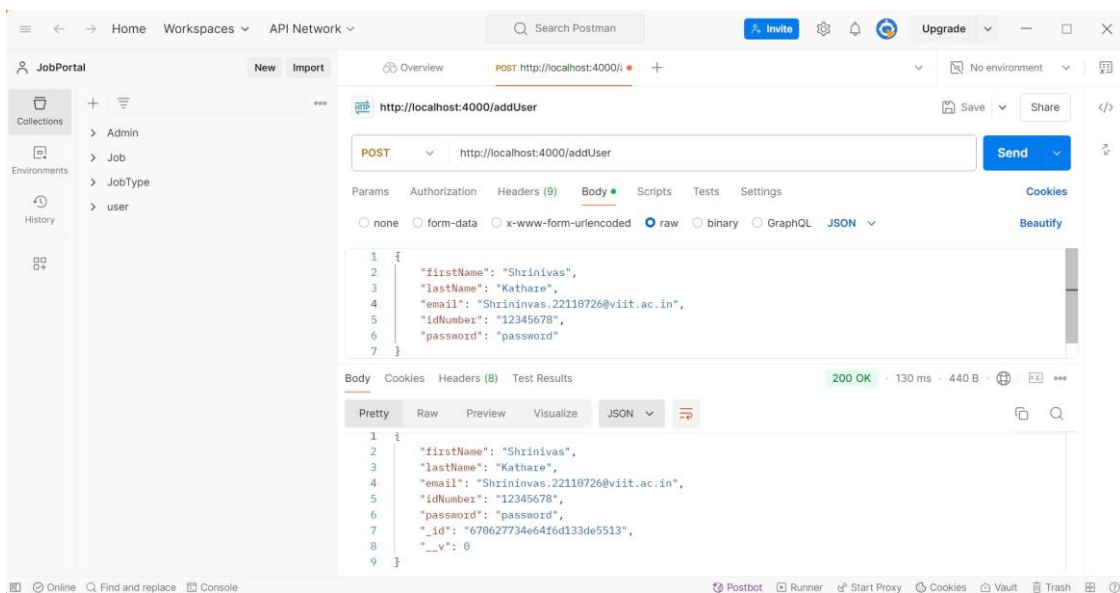


Fig 6.4 Integration Backend-Postman

6.3 TEST CASES AND RESULTS

Test Case No.	Test Case Name	Prerequisites	Action	Expected Result	Actual Result	Status
TC-01	User Registration with Valid Data	None	1. Open registration form 2. Enter valid Aadhar, name, email 3. Submit form	User registered successfully	User registered successfully	Passed
TC-02	User Registration with Invalid Aadhar	None	1. Open registration form 2. Enter invalid Aadhar 3. Submit form	Error message: 'Invalid Aadhar number'	Error message displayed	Passed
TC-03	User Login with Valid Credentials	User registered	1. Open login page 2. Enter valid Aadhar and password 3. Submit form	User logged in successfully	User logged in successfully	Passed
TC-04	User Login with Invalid Credentials	User registered	1. Open login page 2. Enter invalid password 3. Submit form	Error message: 'Invalid credentials'	Error message displayed	Passed
TC-05	Vote Casting with Valid Session	User logged in	1. Log in 2. Navigate to voting page 3. Select candidate 4. Cast vote	Vote casted successfully	Vote casted successfully	Passed
TC-06	User Logout	User logged in	1. Log in 2. Navigate to logout 3. Click logout	User logged out successfully	User logged out successfully	Passed

TC-07	Duplicate User Registration	User already registered	1. Open registration form 2. Enter same Aadhar as an existing user 3. Submit form	Error message: 'User already exists'	Error message displayed	Passed
TC-8	Vote Casting with Invalid Session	User session expired	1. Let session expire 2. Attempt vote	Error message displayed, user must log in again	Error message displayed	Passed
TC-9	Blockchain Record Verification	Admin logged in	1. Access blockchain records 2. Check recent votes	Correct transaction details displayed	Transaction details correct	Passed
TC-10	Unauthorized Admin Access	Regular user logged in	1. Log in as user 2. Try to access admin panel	Error message: 'Access denied'	Error message: 'Access denied'	Passed
TC-11	Dashboard Refresh After Vote	User logged in and vote cast	1. Refresh dashboard after vote	Vote count updates in real-time	Vote count updated	Passed
TC-12	Admin Adds New Candidate	Admin logged in	1. Add new candidate	New candidate added	Candidate added	Passed

Chapter XII

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

With the blockchain-based electronic voting system, voting has significantly advanced, resolving many of the issues with conventional voting procedures. This solution makes sure that the election process is trustworthy and dependable by utilizing the transparent, secure, and immutable properties of blockchain technology. The thorough process described in the project has shown that putting such a system in place may significantly increase voter turnout, reduce the possibility of fraud, and speed up the results while keeping everyone safe. The several test cases that are run verify that the system meets both functional and non-functional criteria and operates as intended. an election environment that is more trustworthy, efficient, and inclusive.

7.2 FUTURE WORK

The blockchain-based electronic voting system offers a plethora of potential for future expansion and improvement. Optimizing performance and scalability is one of the main areas for improvement. It will be crucial to make sure the system can manage a large number of votes effectively without sacrificing response times as the user base grows. This can entail investigating layer-2 approaches to increase scalability and refining the blockchain's underlying protocol. Furthermore, incorporating cutting-edge identity verification techniques like biometric authentication could improve security and guarantee that only qualified voters can take part. Campaigns for public awareness and education will also be essential to the e-voting system's effective implementation. Additionally, continued attempts to engage with politicians will be vital for ensuring that the system conforms with existing laws and adapts to developing legal difficulties.

Chapter XIII

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